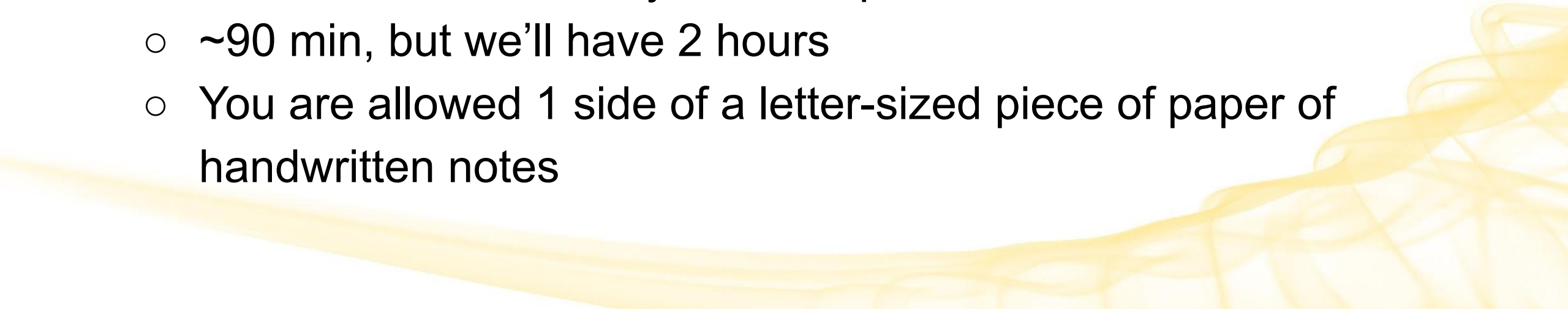


Lecture 27: continuing Deep Learning

CMSC/DATA 320: Introduction to Data Science

Lecturer: Anna Evtushenko (annae@umd.edu)

Announcements

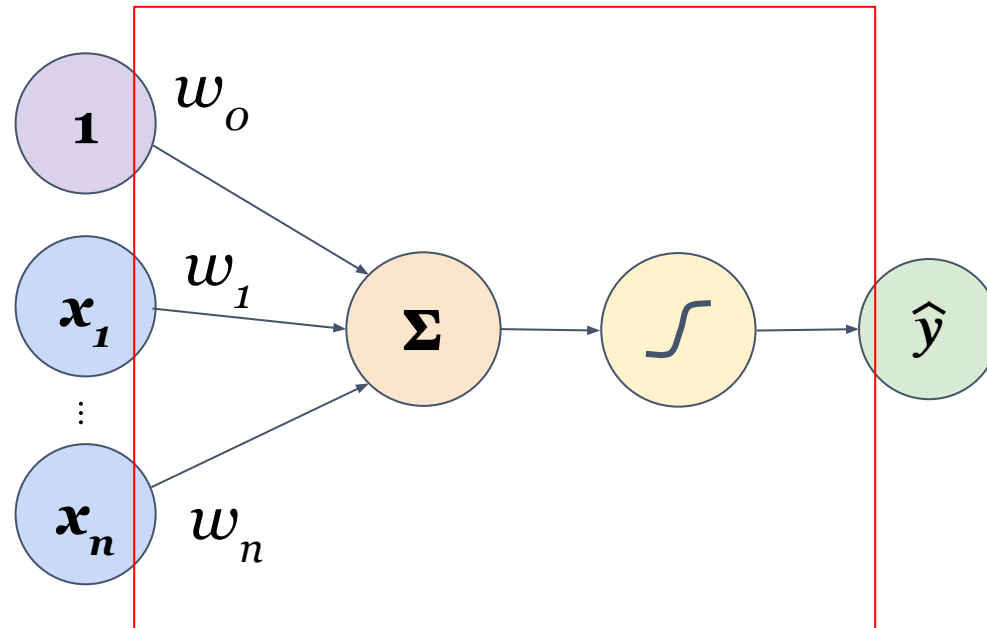
- PA7, WA7 due May 10
 - Last class this Wednesday!
 - Office hours continue (and end) this week!
 - Course evaluations open till May 9!
 - A note about the final
 - Questions similar in style to the quizzes: MCQ and fill in the blank
 - ~90 min, but we'll have 2 hours
 - You are allowed 1 side of a letter-sized piece of paper of handwritten notes
- 

Deep Learning

Deep Learning roadmap

- Overview
 - Neurons
 - Perceptron
 - Neural Networks
 - Feed Forward
 - Backpropagation
- 

A neuron contains

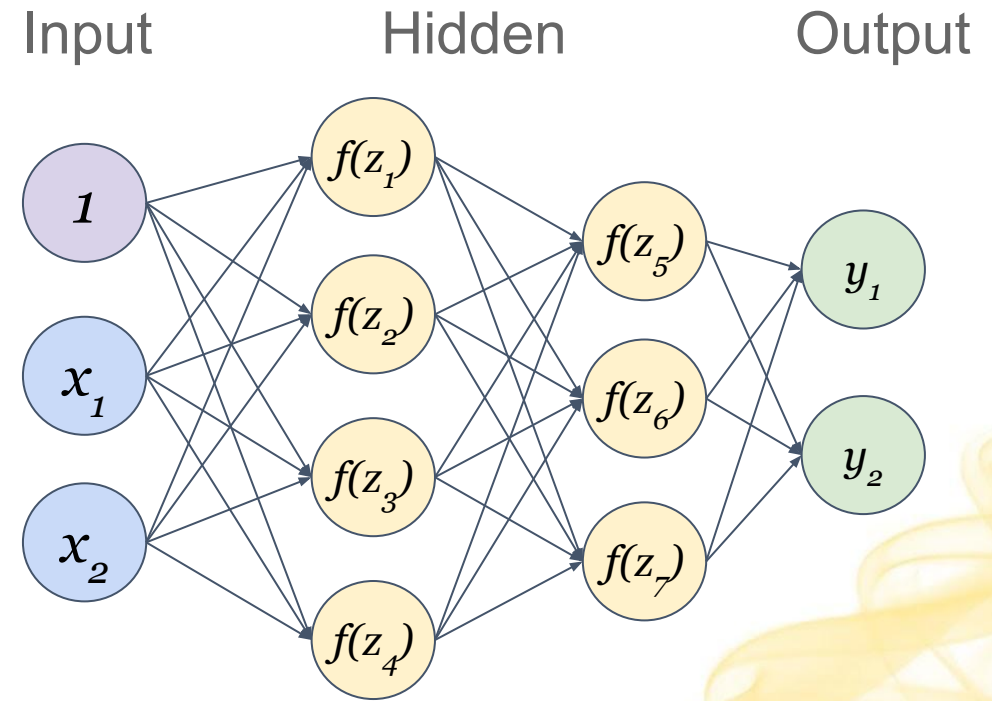


A perceptron model uses one neuron

Multi-Layer Feed-Forward Neural Networks Mechanism

To classify something:

- Activate the input layer with the values of the input vector we want to classify.
- Propagate to each subsequent hidden layer
- Propagate to the output layer for a final prediction.

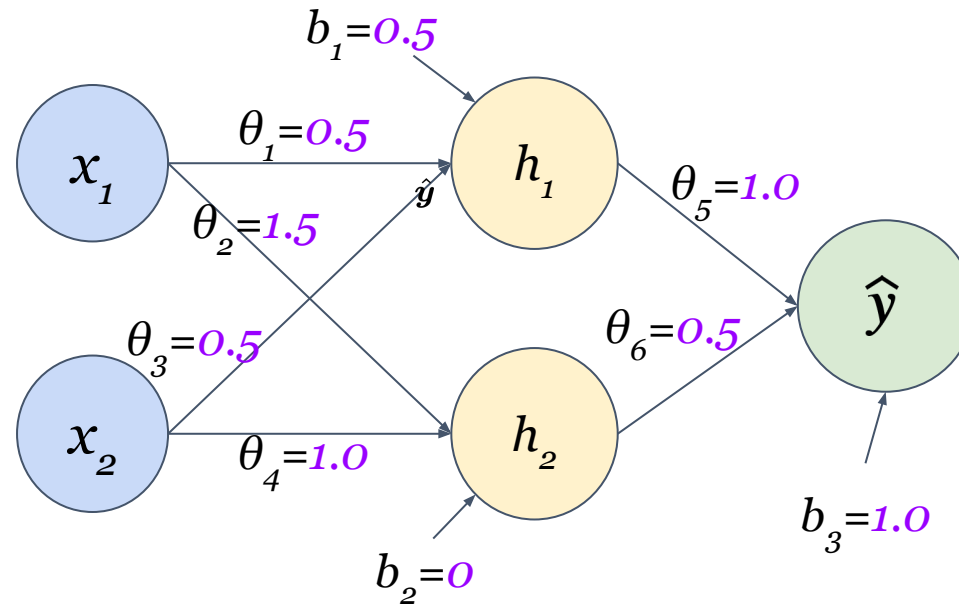


Example

$$x_1=1, x_2=0$$

$$f(z) = \max(0, z) = \begin{cases} z & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

for all non-input nodes

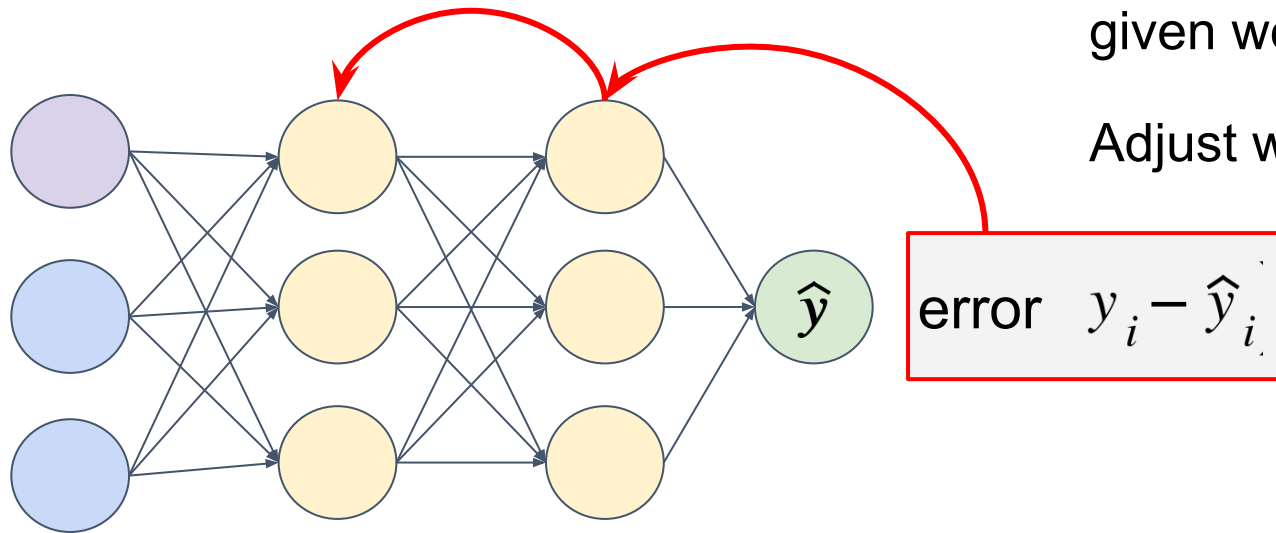


confirm that you get 2.75

Backpropagation



Backpropagation Idea



“Pass the error” backward to neurons by calculating the gradient of the loss function in order to know how much a given weight contributed to the error.

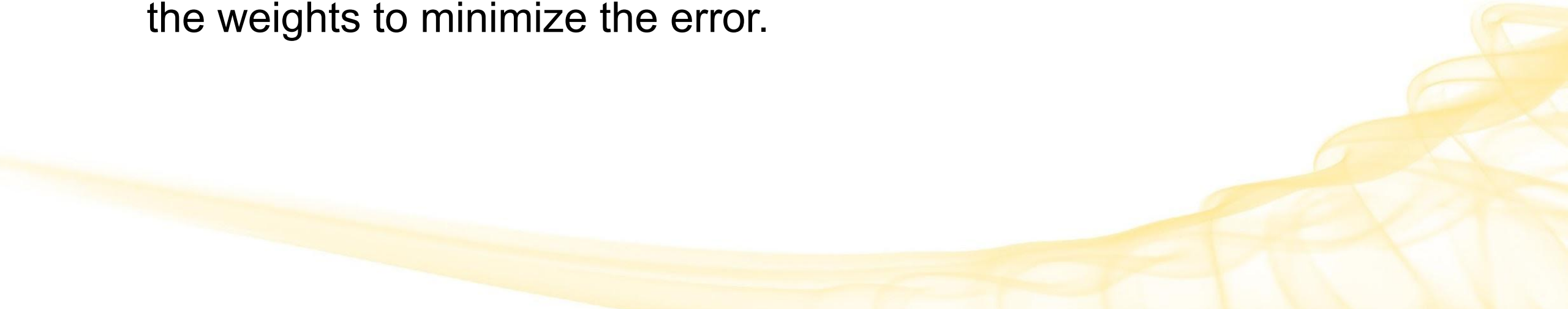
Adjust weights accordingly.

Backpropagation

Backpropagation is the algorithm by which neural networks are trained.

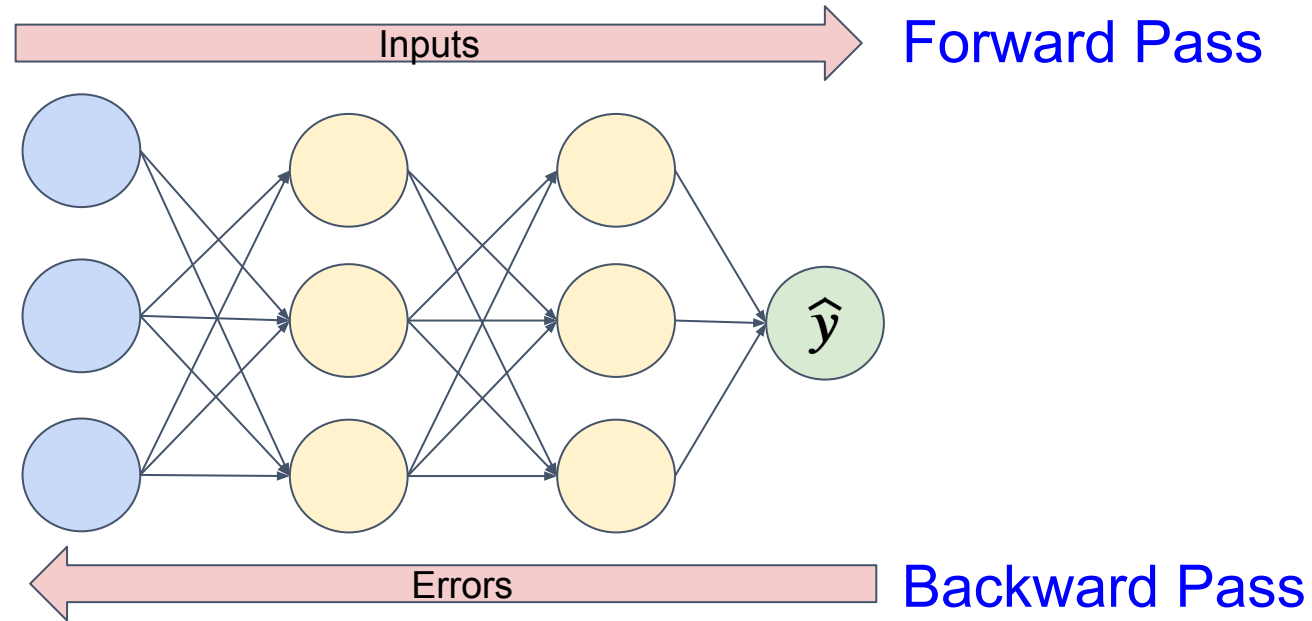
- Backpropagation is short for **Backward Propagation of Errors**
 - Errors are propagated in the opposite direction of the network
- **Goal:** Update weights in the network by minimizing the error (cost function) for each output neuron, and the network as a whole.
- **Why?** To improve output predictions.
- **How?** For every training example, compute the loss function, then iteratively adjust each weight (including bias) based on the gradient of the loss function, which indicates how much each parameter contributed to the prediction error.

Backpropagation Steps

1. Complete the forward pass.
 2. Calculate the loss.
 3. Propagate the error backward through the network.
 4. Using the chain rule to calculate the **contribution of each weight** to the overall error.
 5. Take the derivative of the error with respect to each weight, then adjust the weights to minimize the error.
- 

Overall Process

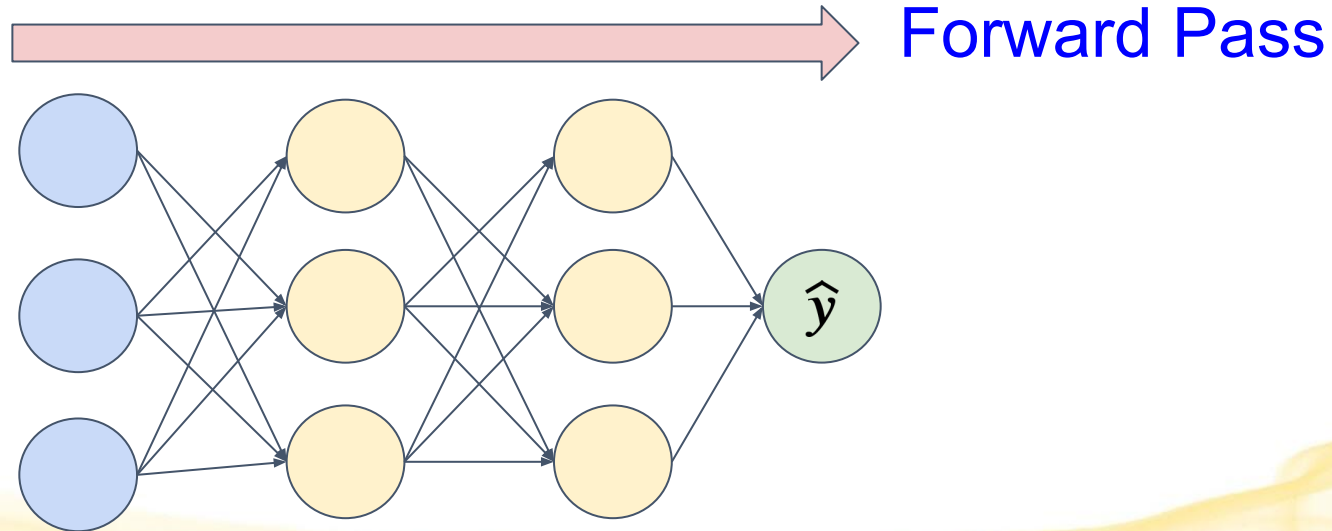
1. Forward Pass (Forward Propagation & Loss Calculation)
2. Backward Pass (Gradient Calculation or Backpropagation)



Backpropagation: Forward Pass

Step 1.1: Forward Propagation: The input data is passed **forward** through the network, producing an output prediction.

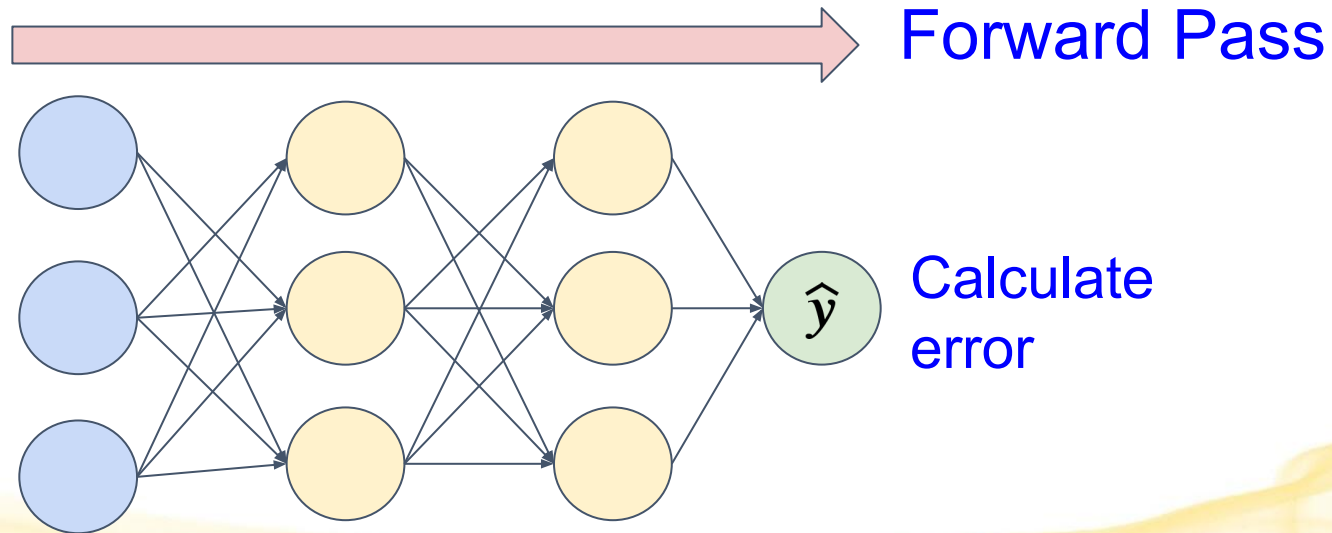
- Each layer computes a weighted sum of inputs and applies an activation function.
- Outputs are generated layer by layer until the final prediction is obtained.



Backpropagation: Calculate Error

Step 1.2: Calculate Loss / Error: Quantify the difference between predicted and actual values.

- Compare the predicted output to the expected target to compute the error, typically using a loss function such as Mean Square Error (MSE).



Cost/Loss Functions

Represent the total loss over our entire network.

Regression

- Same cost function as Linear Regression (MSE) $J(\theta) = MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

Classification

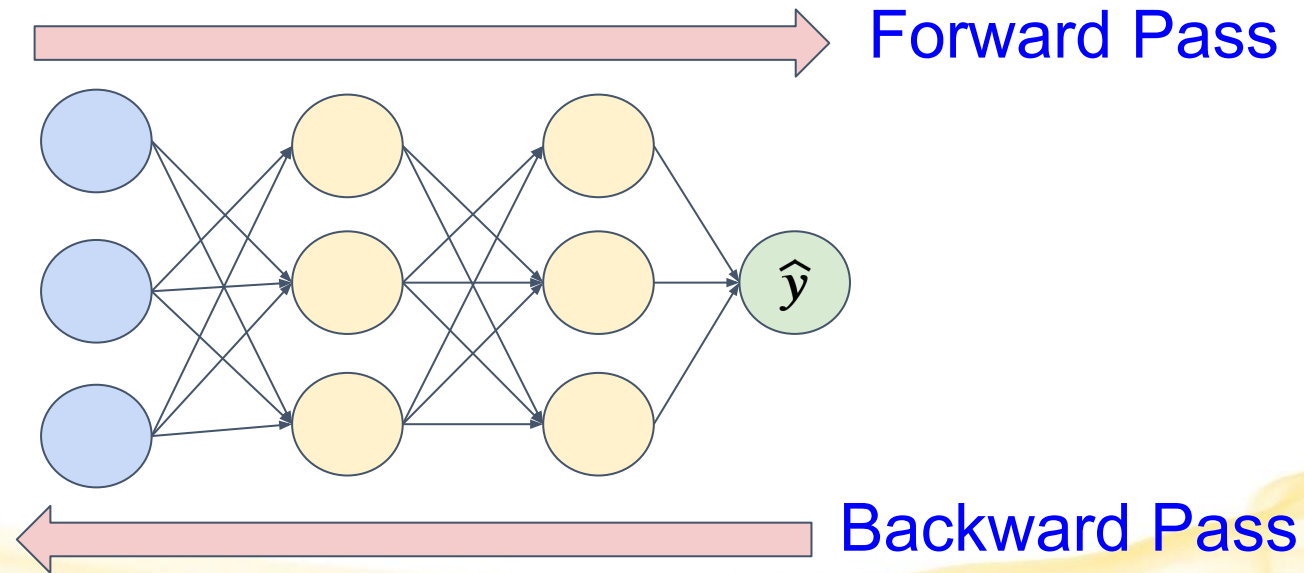
- Same cost function as Logistic Regression (negative natural log of the Logistic/Sigmoid Function, a.k.a. Negative Log Likelihood)
- Based on probabilities (output $\hat{y}$ needs to be on $[0, 1]$)

$$J(\theta) = -\frac{1}{n} \sum Y_i \cdot \ln(h_{\theta}(x_i)) + (1 - Y_i) \cdot \ln(1 - h_{\theta}(x_i))$$

Backpropagation: Backward Pass

Step 2: Backpropagate

- The error is propagated backward through the network.
- Adjust the network's internal parameters (weights, including biases) based on the calculated error, aiming to minimize the difference between the predicted and actual output.



Backpropagation Cost Minimization

Say we have the following loss function: $J(\theta) = MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

What we would like to know is

$$\frac{\partial J}{\partial \theta_j}$$

Indicates the rate of change (slope) of the loss, with respect to a particular weight estimate, θ_j .

For all of the different weights in the network, weights with a **high partial derivative** strongly contributed to the error.

Conversely, weights with a **low partial derivative** had less influence on the error.

Backpropagation Cost Minimization

We now use an iterative optimization algorithm like Gradient Descent in order to optimize (minimize) our cost function using the calculated partial derivatives for every node in the network.

Just as before, we **update** our weight estimates at each iteration:

$$\theta_{new} = \theta - \alpha \cdot \nabla J(\theta) :$$

$$\theta_0 = \theta_0 - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_0}$$
$$\vdots$$

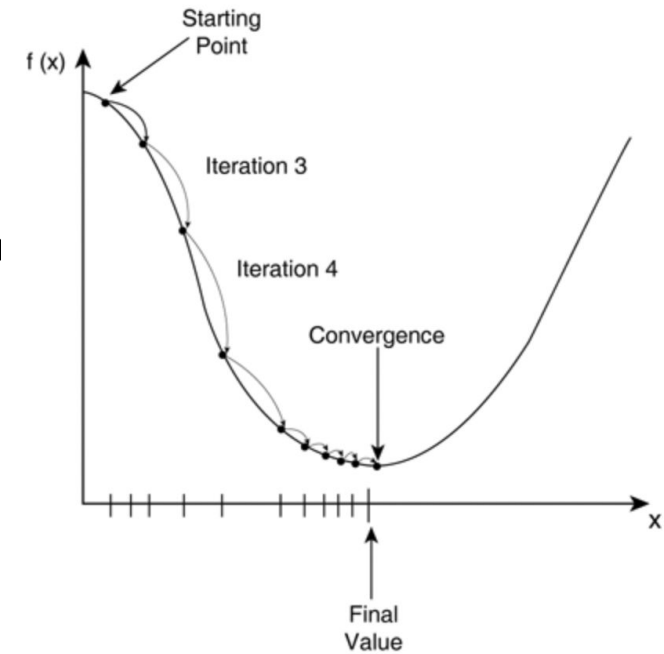
$$\theta_j = \theta_j - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_j}$$

α is the Learning Rate
(step size), just as
before.

Backpropagation w/ Cost Optimization

Backpropagation, along with an optimization algorithm like Gradient Descent, forms the **backbone of neural networks**.

- **Backpropagation** calculates the derivatives (evaluates the gradient of the cost) with respect to the weights
 - Gives us the magnitude and direction for Gradient Descent to use, and tells it which weights need adjusting.
- **Gradient Descent** descends through the gradient
 - Iteratively updates and moves the weights towards the minimum of the cost function.

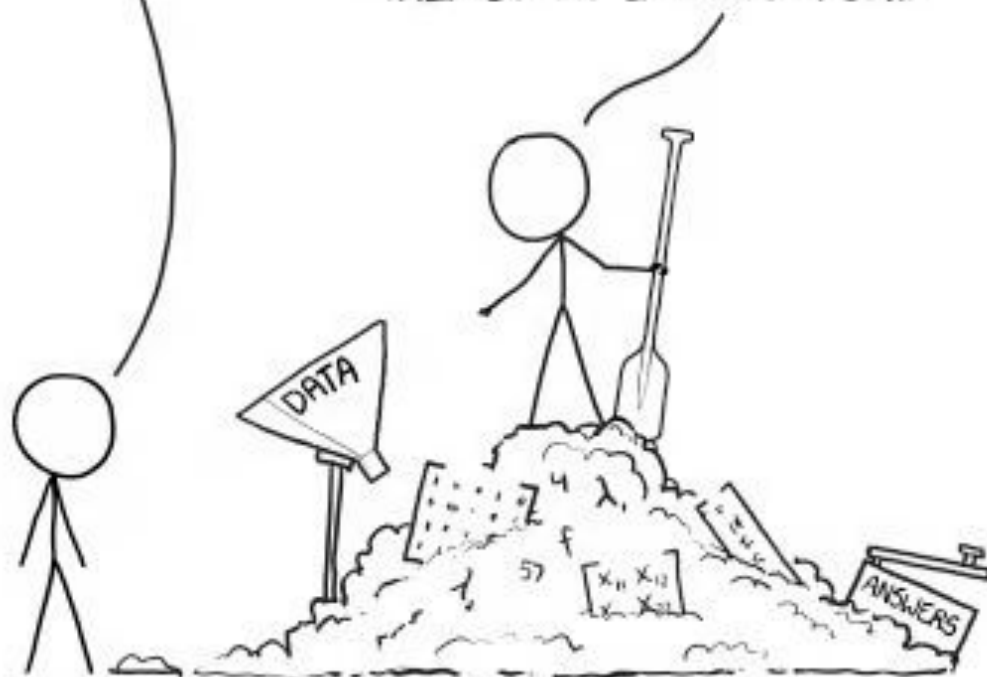


THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG
PILE OF LINEAR ALGEBRA, THEN COLLECT
THE ANSWERS ON THE OTHER SIDE.

WHAT IF THE ANSWERS ARE WRONG?

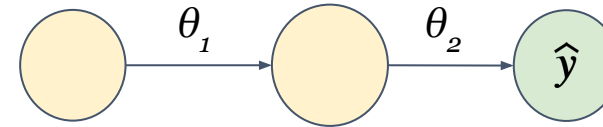
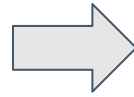
JUST STIR THE PILE UNTIL
THEY START LOOKING RIGHT.



Chain Rule in Backpropagation

We can use the simple chain rule from calculus and apply it to neural network backpropagation.

$$\frac{d}{dx} f(g(x)) = \frac{df}{dg} \cdot \frac{dg}{dx}$$



$$\frac{\delta J(\theta)}{\delta \theta_2} = \frac{\delta J(\theta)}{\delta \hat{y}} \cdot \frac{\delta \hat{y}}{\delta \theta_2}$$

$$\frac{\delta J(\theta)}{\delta \theta_1} = \frac{\delta J(\theta)}{\delta \hat{y}} \cdot \frac{\delta \hat{y}}{\delta \theta_1} = \frac{\delta J(\theta)}{\delta \hat{y}} \cdot \frac{\delta \hat{y}}{\delta f(z)} \cdot \frac{\delta f(z)}{\delta \theta_1}$$

Do this for every node.

Downstream
Gradient Upstream
Gradient Local
Gradient

The chain rule intuition

The Chain Rule allows us to calculate the influence of a "deep" weight by multiplying local connections.

If A affects B, and B affects C, how does A affect C?

Mathematically, the Chain Rule says:
$$\frac{\partial C}{\partial A} = \frac{\partial C}{\partial B} \cdot \frac{\partial B}{\partial A}$$

In backpropagation, it looks like this:

We calculate how much the Error changes relative to the Output.

We calculate how much the Output changes relative to the Hidden Layer.

We multiply them together to find out how much the Error changes relative to the Hidden Layer.

“Simple” Example

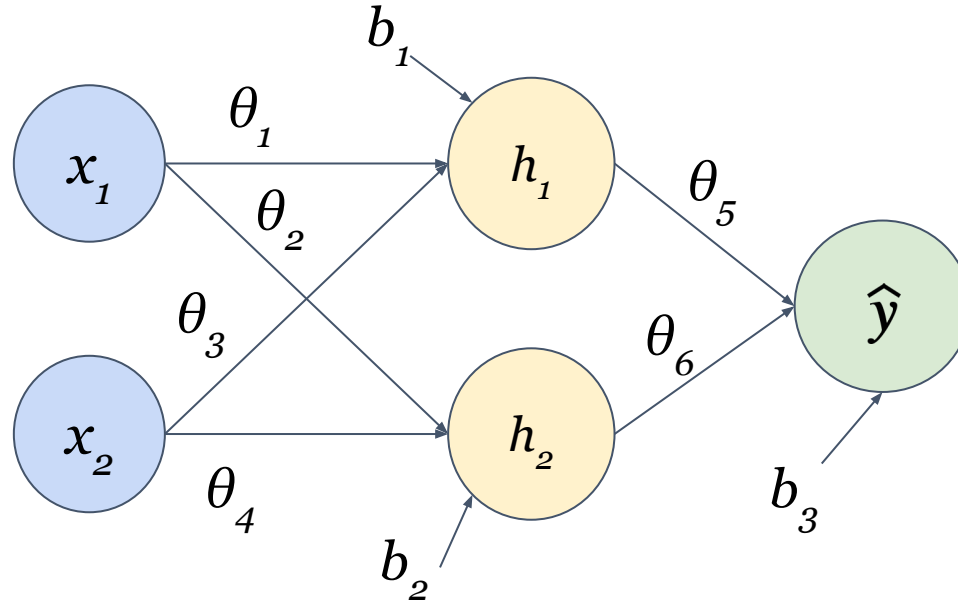
Suppose we have a network that we want to train to learn the function

$$y = x_1^2 + x_2 + 1$$

We'll use ReLU for our activation function for simplicity.

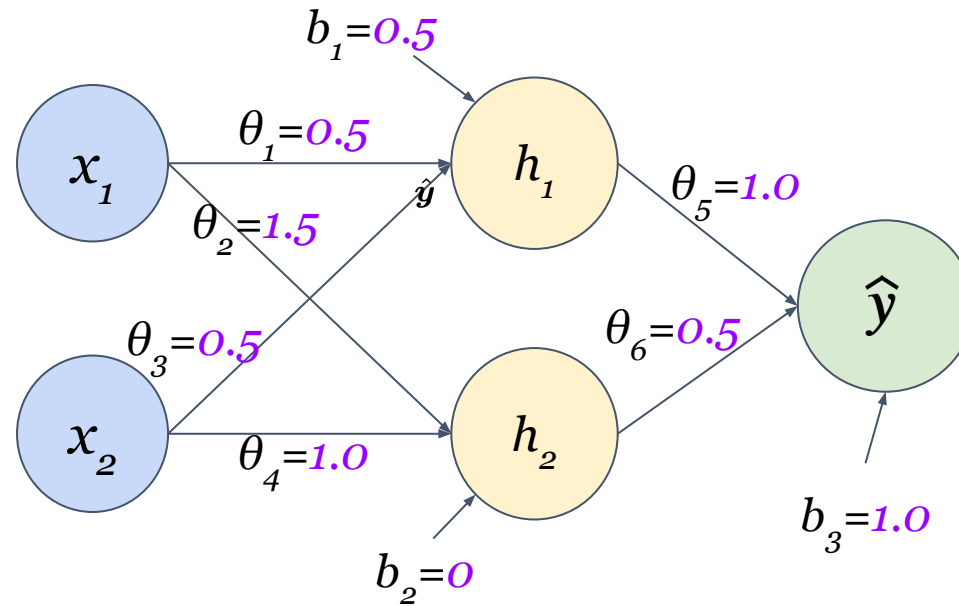
$$f(z) = \max(0, z) = \begin{cases} z & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

And a learning rate of $\alpha=0.1$



Simple Example

First, we initialize the weights and biases to random numbers.



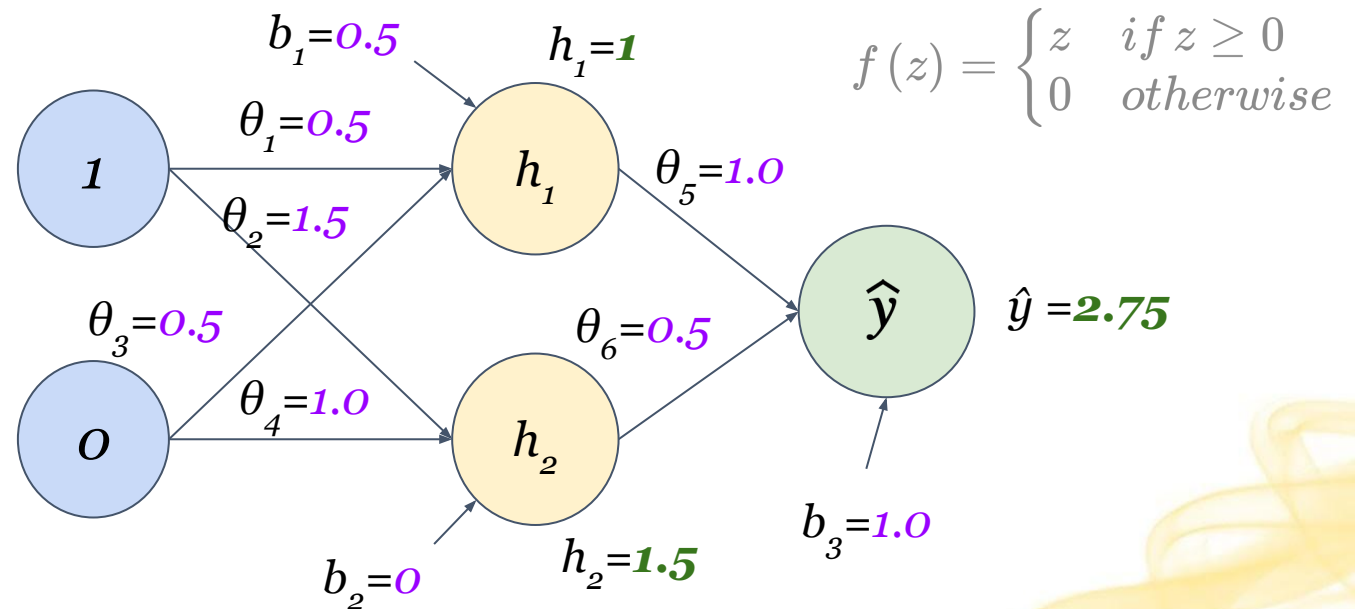
Simple Example

Now we bring in a training observation ($x_1=1, x_2=0, y=2$) and make a **Forward Pass**.

$$\begin{aligned}h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \\&= f(0.5 + 0.5*1 + 0.5*0) = f(1) = 1\end{aligned}$$

$$\begin{aligned}h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \\&= f(0 + 1.5*1 + 1*0) = f(1.5) = 1.5\end{aligned}$$

$$\begin{aligned}\hat{y} &= f(b_3 + \theta_5 h_1 + \theta_6 h_2) \\&= f(1 + 1*1 + 0.5*1.5) = f(2.75) = 2.75\end{aligned}$$



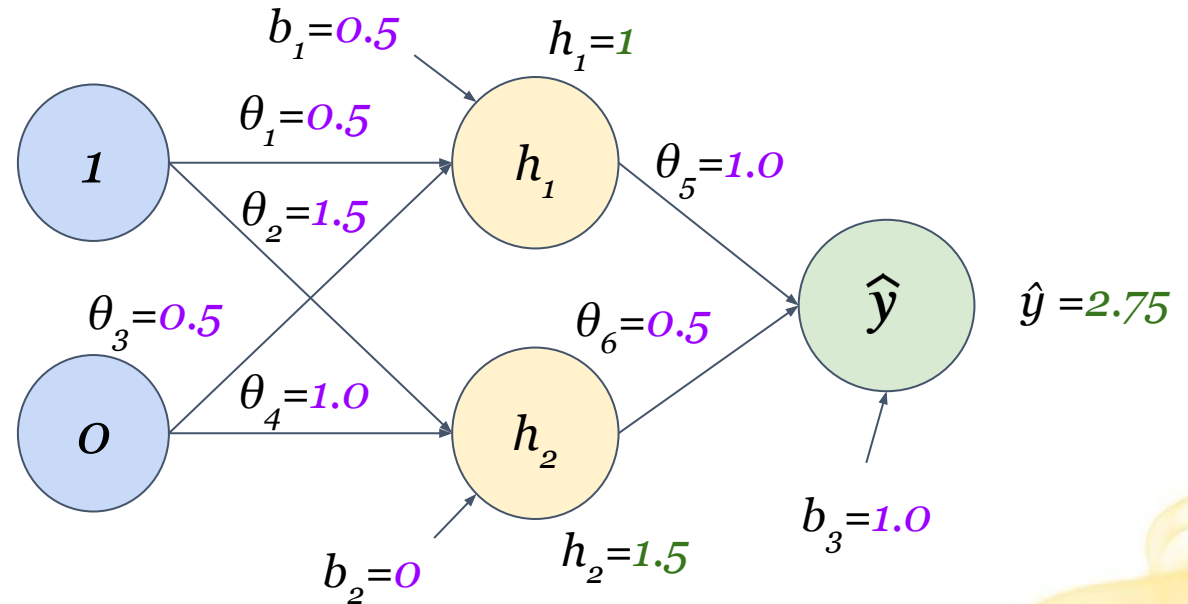
So: $h_1=1, h_2=1.5, \hat{y}=2.75$

Simple Example

Next, we calculate the error for our prediction on observation ($x_1=1, x_2=0, y=2$).

$$h_1=1, h_2=1.5, \hat{y}=2.75$$

$$\begin{aligned} \text{Squared Error} &= (y - \hat{y})^2 \\ &= (2 - 2.75)^2 = -0.75^2 = \mathbf{0.5625} \end{aligned}$$



Simple Example

Now we make a **Backward Pass**.

We'll first take the partial derivative of the cost function with respect to estimated weight θ_6 .

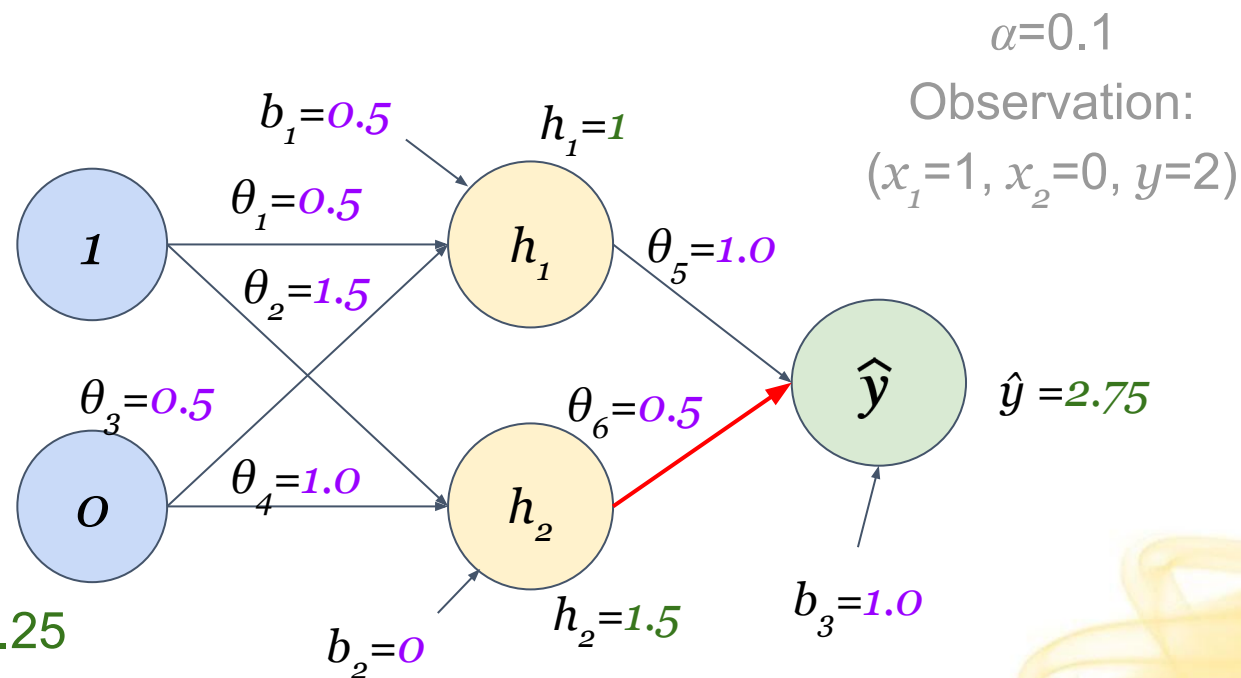
For simplicity, suppose we don't use an activation function on the output (we'll look at f in a minute).

$$\begin{aligned}\frac{\partial J}{\partial \theta_6} &= \frac{\partial (y - \hat{y})^2}{\partial \theta_6} = \frac{\partial (y - b_3 - \theta_5 h_1 - \theta_6 h_2)^2}{\partial \theta_6} \\ &= -2h_2 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) \\ &= -2 * 1.5 * (2 - 1 - 1 * 1 - 0.5 * 1.5) = -3(0 - 0.75) = 2.25\end{aligned}$$

Then update θ_6 :

$$\theta_6 = \theta_6 - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_6} = 0.5 - 0.1 * 2.25 = 0.5 - 0.225 = 0.275$$

$$\begin{aligned}h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75\end{aligned}$$



Simple Example

Now we make a **Backward Pass**.

We'll first take the partial derivative of the cost function with respect to estimated weight θ_6 .

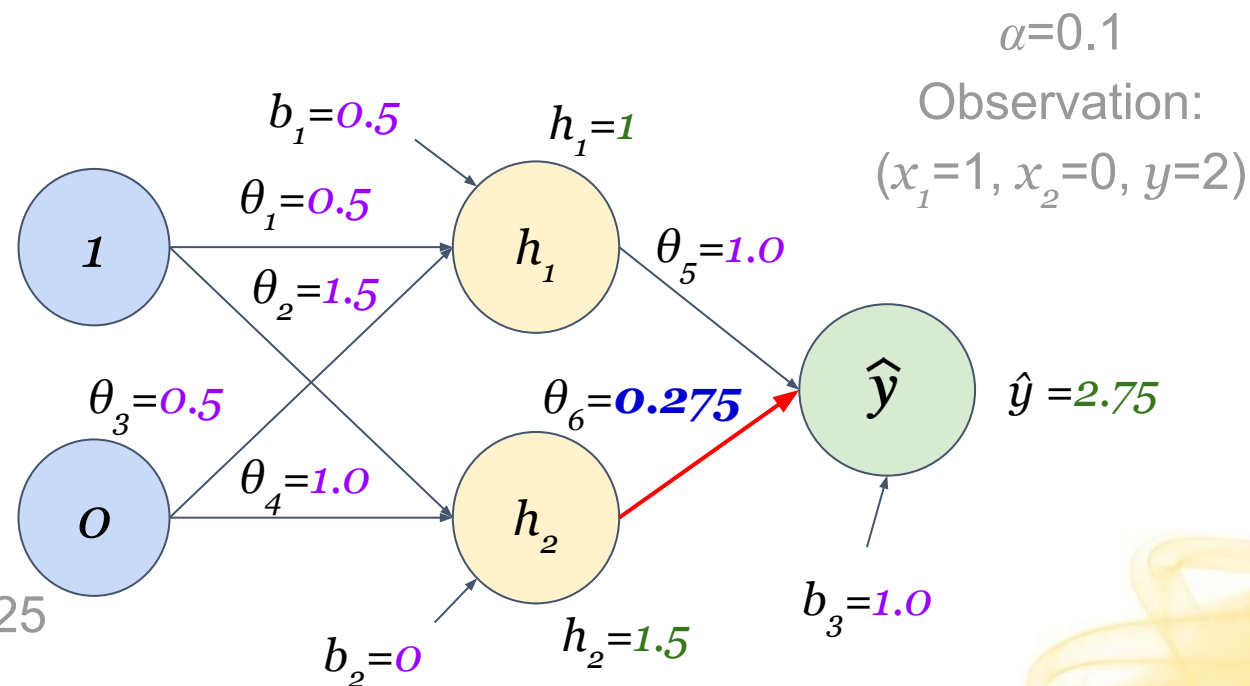
For simplicity, suppose we don't use an activation function on the output (will look at f in a minute).

$$\begin{aligned}\frac{\partial J}{\partial \theta_6} &= \frac{\partial (y - \hat{y})^2}{\partial \theta_6} = \frac{\partial (y - b_3 - \theta_5 h_1 - \theta_6 h_2)^2}{\partial \theta_6} \\ &= 2h_2 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) \\ &= 2 \cdot -1.5 \cdot (2 - 1 - 1 \cdot 1 - 0.5 \cdot 1.5) = -3(0 - 0.75) = 2.25\end{aligned}$$

Then update θ_6 :

$$\theta_6 = \theta_6 - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_6} = 0.5 - 0.1 \cdot 2.25 = 0.5 - 0.225 = \mathbf{0.275}$$

$$\begin{aligned}h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75\end{aligned}$$



Simple Example

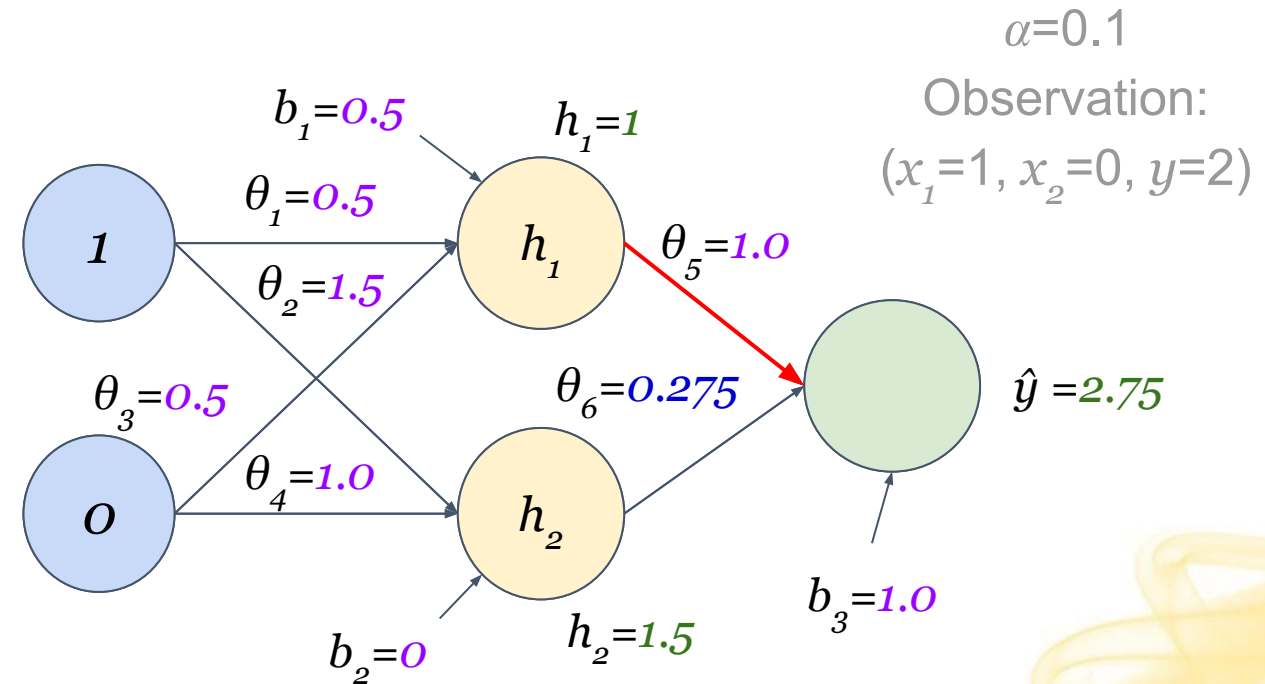
$$\begin{aligned}h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1=1 \\h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2=1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y}=2.75\end{aligned}$$

We can do the same for θ_5 .

$$\begin{aligned}\frac{\partial J}{\partial \theta_5} &= \frac{\partial (y - \hat{y})^2}{\partial \theta_5} = \frac{\partial (y - b_3 - \theta_5 h_1 - \theta_6 h_2)^2}{\partial \theta_5} \\&= -2h_1 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) \\&= -2 \cdot 1 (2 - 1 - 1 \cdot 1 - 0.5 \cdot 1.5) = -2(0 - 0.75) \\&= 1.5\end{aligned}$$

Then update θ_5 :

$$\theta_5 = \theta_5 - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_5} = 1 - 0.1 \cdot 1.5 = 1 - 0.15 = 0.85$$



Simple Example

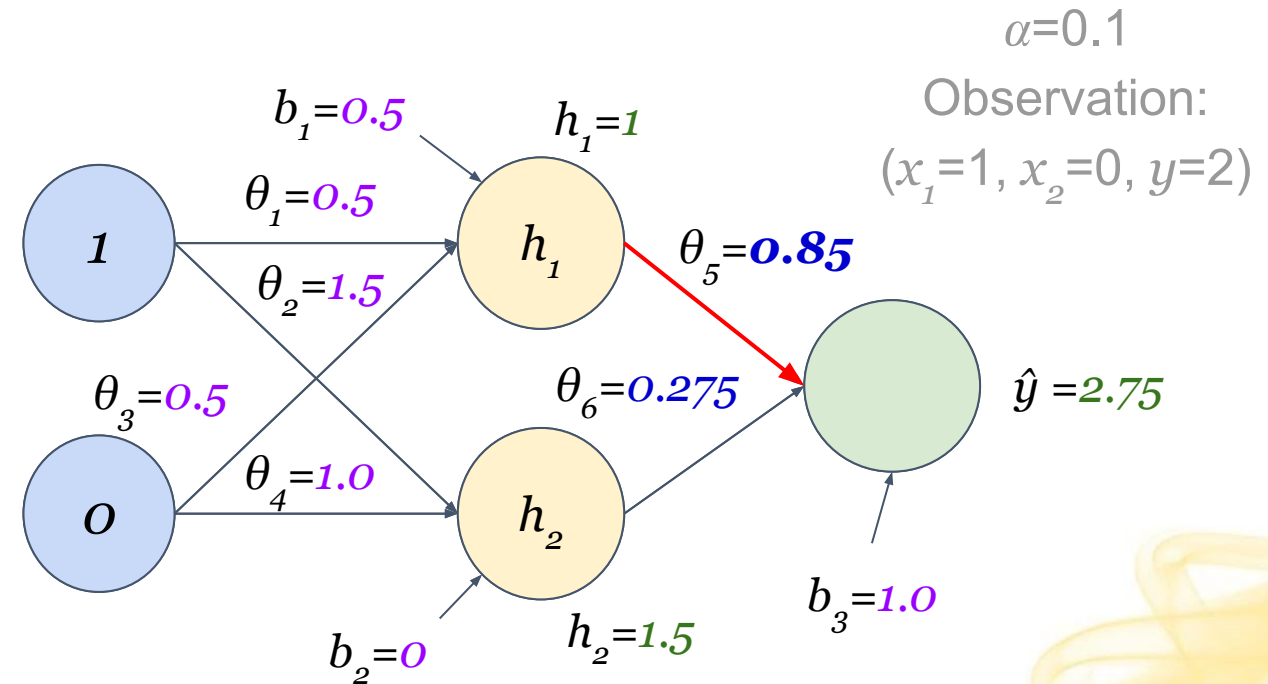
$$\begin{aligned}h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1=1 \\h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2=1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y}=2.75\end{aligned}$$

We can do the same for θ_5 .

$$\begin{aligned}\frac{\partial J}{\partial \theta_5} &= \frac{\partial (y - \hat{y})^2}{\partial \theta_5} = \frac{\partial (y - b_3 - \theta_5 h_1 - \theta_6 h_2)^2}{\partial \theta_5} \\&= -2h_1 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) \\&= -2 \cdot 1 (2 - 1 - 1 \cdot 1 - 0.5 \cdot 1.5) = -2(0 - 0.75) \\&= 1.5\end{aligned}$$

Then update θ_5 :

$$\theta_5 = \theta_5 - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_5} = 1 - 0.1 \cdot 1.5 = 1 - 0.15 = \mathbf{0.85}$$



Simple Example

$$h_1 = f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1$$

$$h_2 = f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5$$

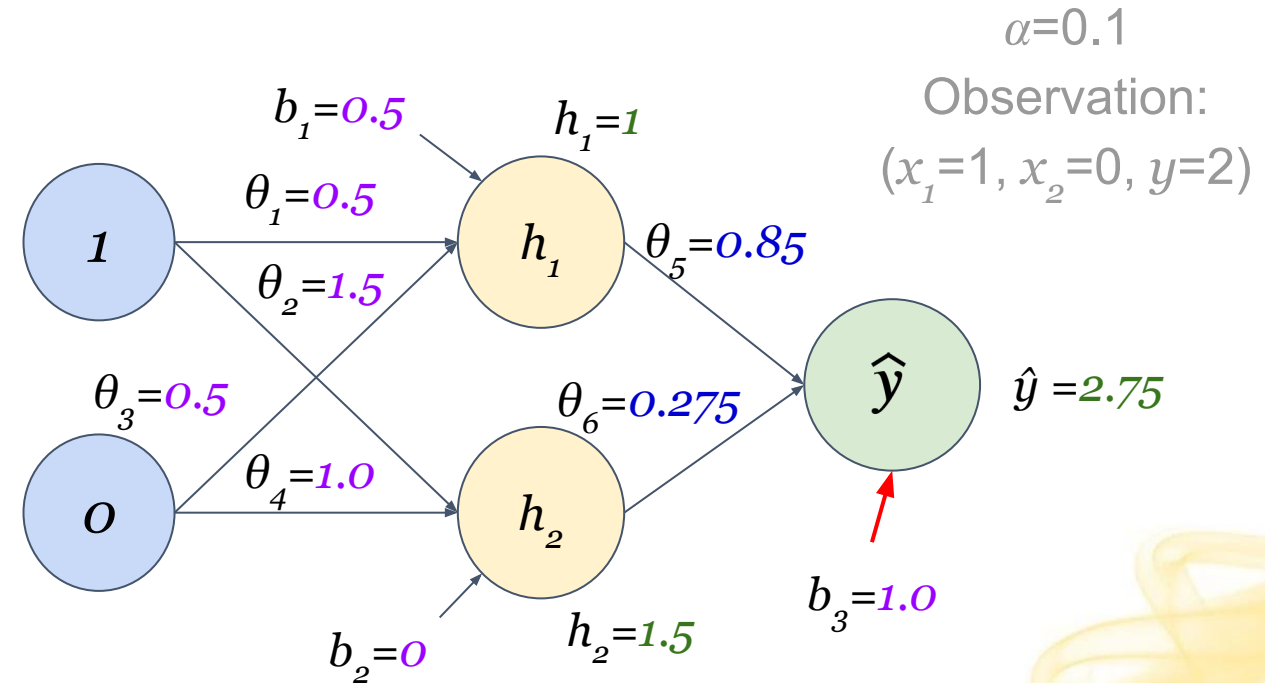
$$\hat{y} = b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75$$

Same for b_3 :

$$\begin{aligned} \frac{\partial J}{\partial b_3} &= \frac{\partial (y - \hat{y})^2}{\partial b_3} = \frac{\partial (y - b_3 - \theta_5 h_1 - \theta_6 h_2)^2}{\partial b_3} \\ &= -2 \cdot 1 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) \\ &= -2(2 - 1 - 1 \cdot 1 - 0.5 \cdot 1.5) = -2(0 - 0.75) \\ &= 1.5 \end{aligned}$$

Then update b_3 :

$$b_3 = b_3 - \alpha \cdot \frac{\partial J(\theta)}{\partial b_3} = 1 - 0.1 \cdot 1.5 = 1 - 0.15 = 0.85$$



Simple Example

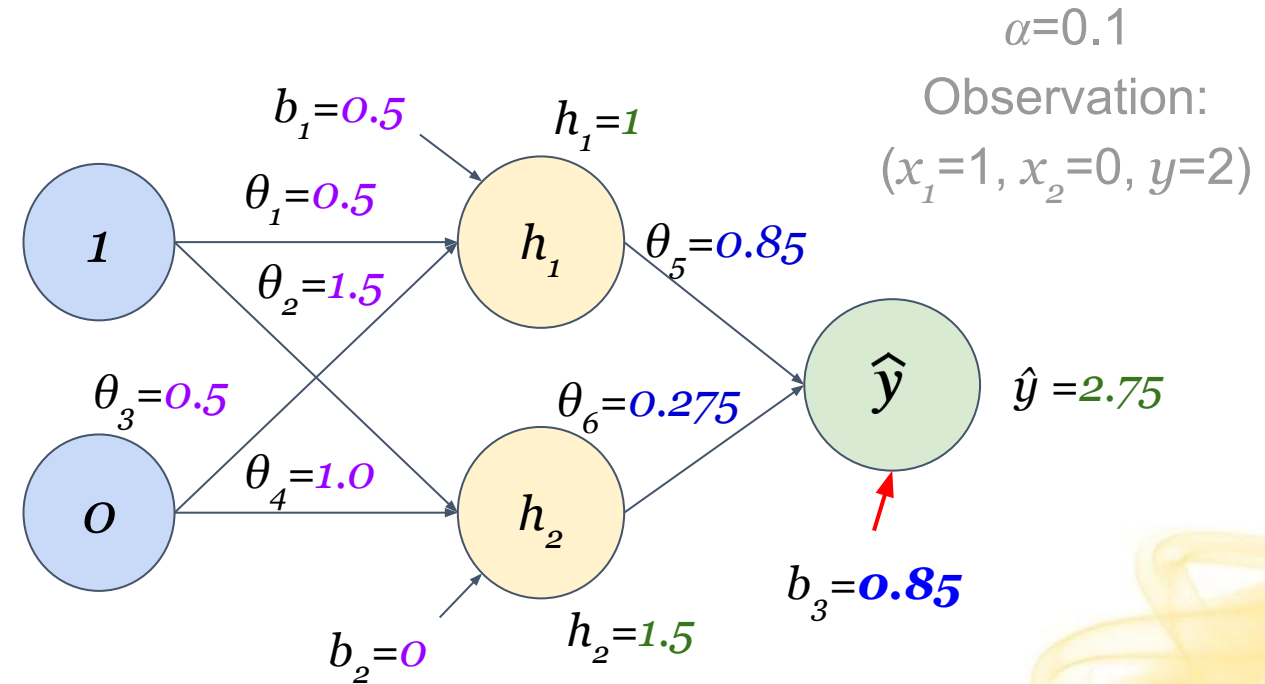
$$\begin{aligned}h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75\end{aligned}$$

Same for b_3 .

$$\begin{aligned}\frac{\partial J}{\partial b_3} &= \frac{\partial (y - \hat{y})^2}{\partial b_3} = \frac{\partial (y - b_3 - \theta_5 h_1 - \theta_6 h_2)^2}{\partial b_3} \\&= -2 \cdot 1 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) \\&= -2(2 - 1 - 1 \cdot 1 - 0.5 \cdot 1.5) = -2(0 - 0.75) \\&= 1.5\end{aligned}$$

Then update b_3 :

$$b_3 = b_3 - \alpha \cdot \frac{\partial J(\theta)}{\partial b_3} = 1 - 0.1 \cdot 1.5 = 1 - 0.15 = \mathbf{0.85}$$



Simple Example

Now go back a layer, to θ_4 .

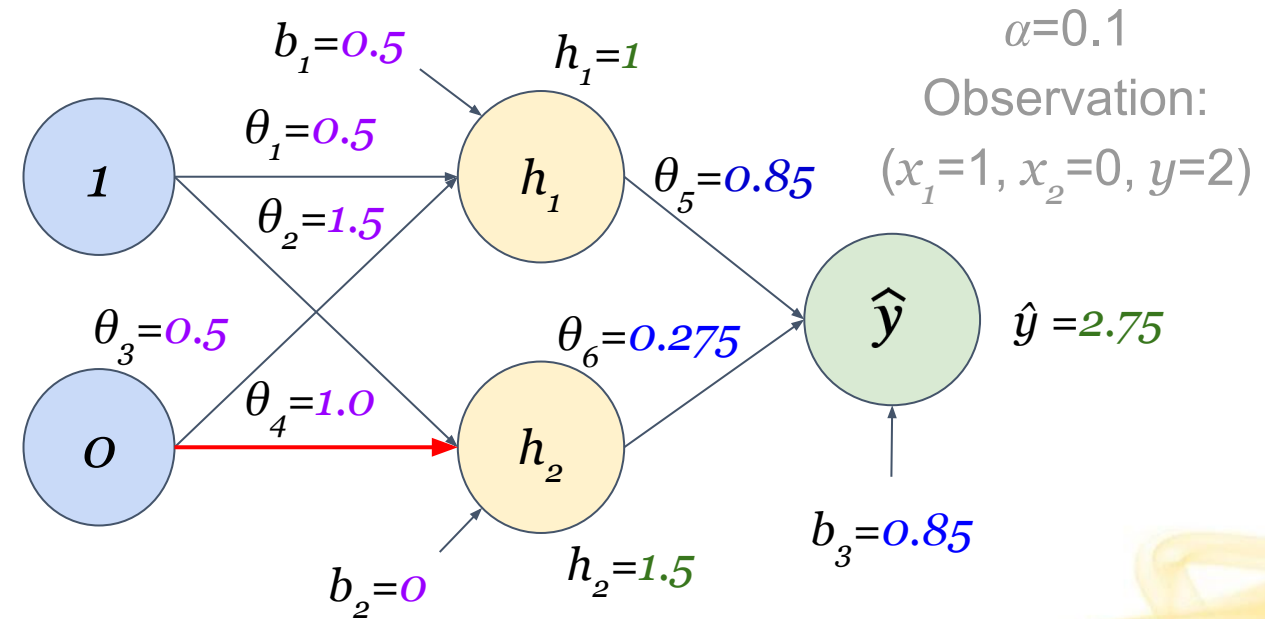
In this case, we need to use the chain rule to expand our cost function, because in order to take the partial derivative with respect to θ_4 we'll need to expand h_2 to its function which includes θ_4 .

$$\frac{\partial J}{\partial \theta_4} = \frac{\partial (y - \hat{y})^2}{\partial \theta_4} = \frac{\partial (y - b_3 - \theta_5 h_1 - \theta_6 h_2)^2}{\partial \theta_4}$$

Chain Rule: $\frac{\partial J(\theta)}{\partial \theta_4} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial \theta_4} \Rightarrow \frac{\partial J(\theta)}{\partial h_2} = -2 \cdot \theta_6 (y - b_3 - \theta_5 h_1 - \theta_6 h_2)$

$$= -2 \cdot 0.5 (2 - 1 - 1 \cdot 1 - 0.5 \cdot 1.5) = -1(-0.75) = 0.75$$

$$\begin{aligned} h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75 \end{aligned}$$



Simple Example

Now go back a layer to θ_4 .

$$\frac{\partial J}{\partial \theta_4} = \frac{\partial (y - \hat{y})^2}{\partial \theta_4} = \frac{\partial (y - b_3 - \theta_5 h_1 - \theta_6 h_2)^2}{\partial \theta_4}$$

$$\text{Chain Rule: } \frac{\partial J(\theta)}{\partial \theta_4} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial \theta_4}$$

$$\frac{\partial J(\theta)}{\partial h_2} = -2 \cdot \theta_6 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) = 0.75$$

$$\frac{\partial h_2}{\partial \theta_4} = \frac{\partial (f(b_2 + \theta_2 x_1 + \theta_4 x_2))}{\partial \theta_4}$$

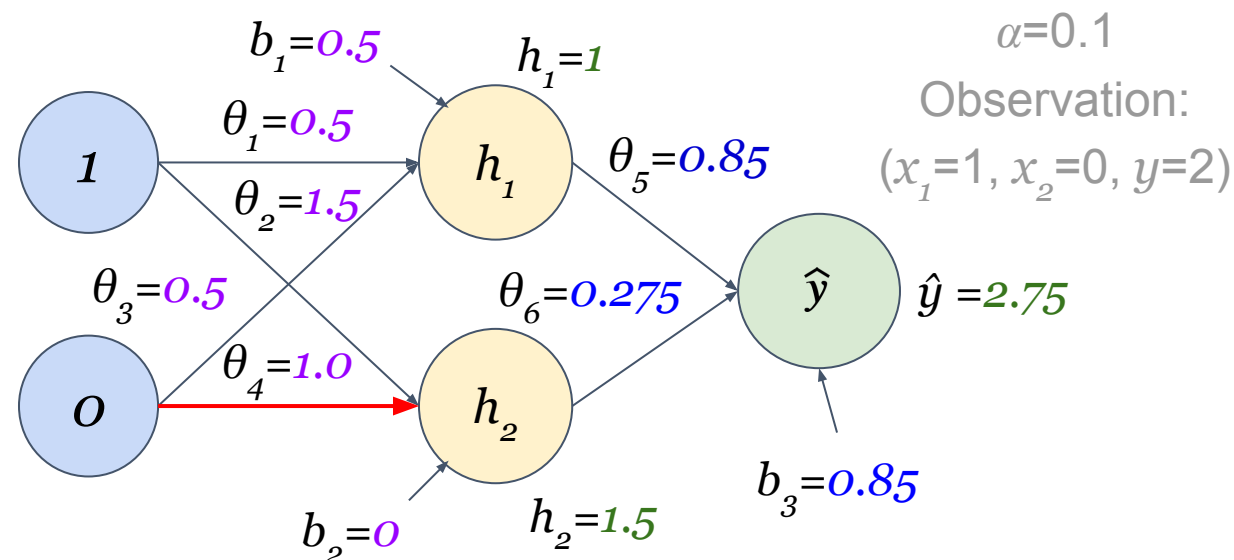
Notice that again we need to use the Chain Rule because θ_4 is inside our activation function f .

$$f(z) = \begin{cases} z & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\frac{\partial h_2}{\partial \theta_4} = \frac{\partial h_2}{\partial z} \cdot \frac{\partial z}{\partial \theta_4} \Rightarrow \frac{\partial h_2}{\partial z} = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \frac{\partial z}{\partial \theta_4} = \frac{\partial (b_2 + \theta_2 x_1 + \theta_4 x_2)}{\partial \theta_4} = x_2 = 0$$

$$\frac{\partial h_2}{\partial z} = \begin{cases} 1 & \text{if } b_2 + \theta_2 x_1 + \theta_4 x_2 \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75 \end{aligned}$$



Simple Example

Now go back a layer to θ_4 .

$$\frac{\partial J(\theta)}{\partial \theta_4} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial \theta_4}$$

$$\frac{\partial J(\theta)}{\partial h_2} = -2 \cdot \theta_6 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) = 0.75$$

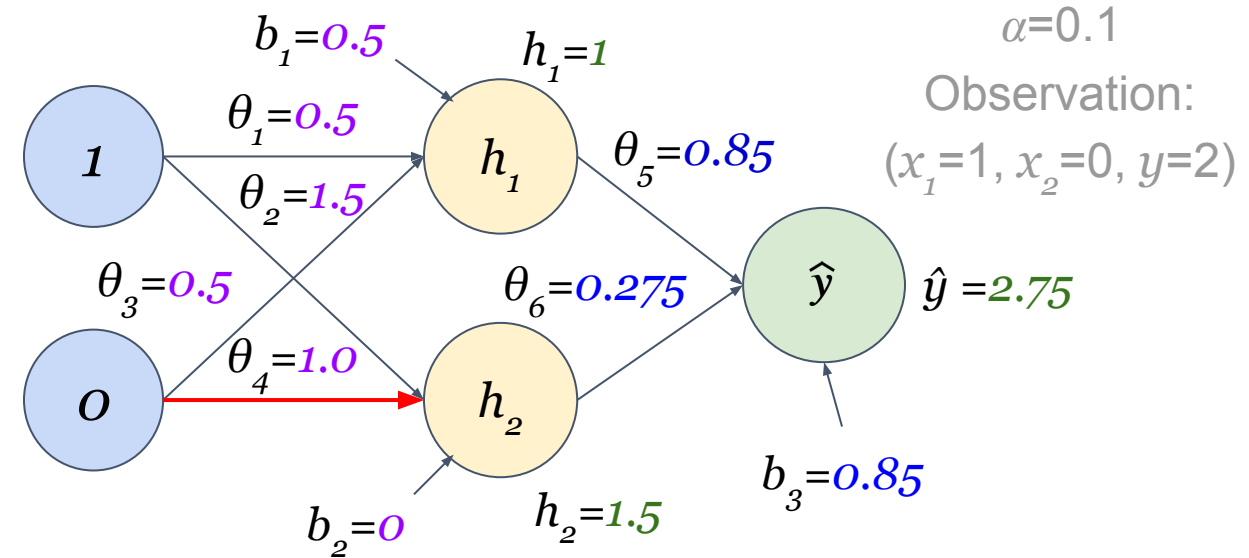
$$\frac{\partial z}{\partial \theta_4} = \frac{\partial (b_2 + \theta_2 x_1 + \theta_4 x_2)}{\partial \theta_4} = x_2 = 0$$

$$\frac{\partial h_2}{\partial z} = \begin{cases} 1 & \text{if } b_2 + \theta_2 x_1 + \theta_4 x_2 \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Now plug in: $b_2 + \theta_2 x_1 + \theta_4 x_2 = 0 + 1.5 \cdot 1 + 1 \cdot 0 = 1.5 > 0 \Rightarrow \frac{\partial h_2}{\partial z} = 1.5$

$$\frac{\partial J(\theta)}{\partial \theta_4} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial \theta_4} = 0.75 \cdot 0 \cdot 1.5 = 0$$

$$\begin{aligned} h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75 \end{aligned}$$



Simple Example

Now go back a layer to θ_4 .

$$\frac{\partial J(\theta)}{\partial \theta_4} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial \theta_4}$$

$$\frac{\partial J(\theta)}{\partial h_2} = -2 \cdot \theta_6 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) = 0.75$$

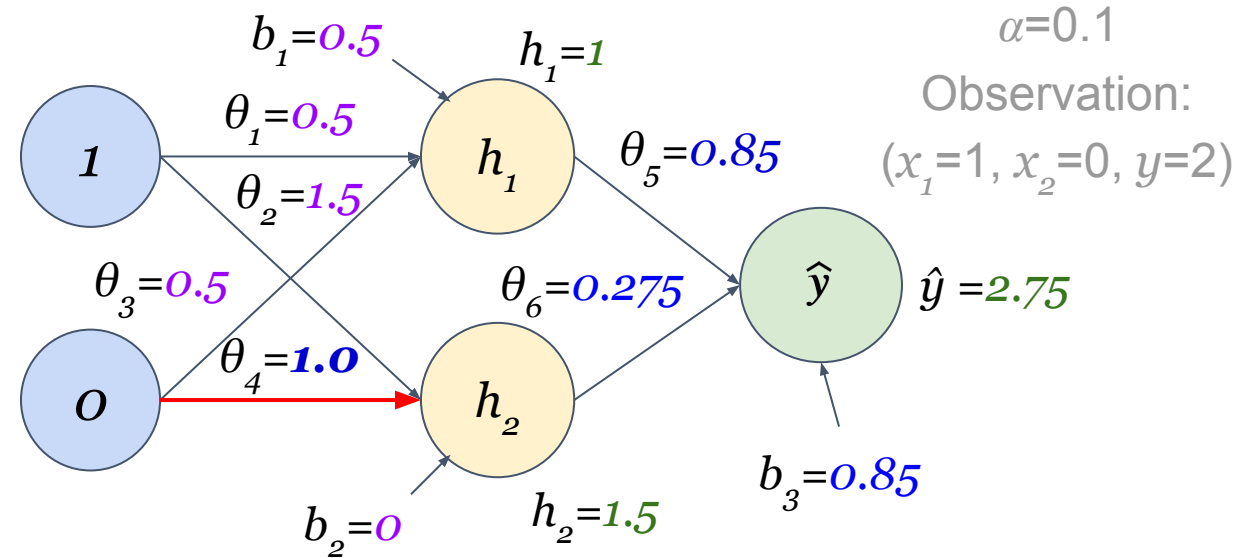
$$\frac{\partial z}{\partial \theta_4} = \frac{\partial (b_2 + \theta_2 x_1 + \theta_4 x_2)}{\partial \theta_4} = x_2 = 0$$

$$\frac{\partial h_2}{\partial z} = \begin{cases} 1 & \text{if } b_2 + \theta_2 x_1 + \theta_4 x_2 \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Now plug in: $b_2 + \theta_2 x_1 + \theta_4 x_2 = 0 + 1.5 \cdot 1 + 1 \cdot 0 = 1.5 > 0 \Rightarrow \frac{\partial h_2}{\partial z} = 1.5$

$$\frac{\partial J(\theta)}{\partial \theta_4} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial \theta_4} = 0.75 \cdot 0 \cdot 1.5 = 0$$

$$\begin{aligned} h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75 \end{aligned}$$



Now update θ_4 :

$$\theta_4 = \theta_4 - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_4}$$

$$= 1 - 0.1 \cdot 0 = 1 \rightarrow \text{no change}$$

Simple Example

We can do the same thing for θ_2 .

$$\frac{\partial J(\theta)}{\partial \theta_2} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial \theta_2}$$

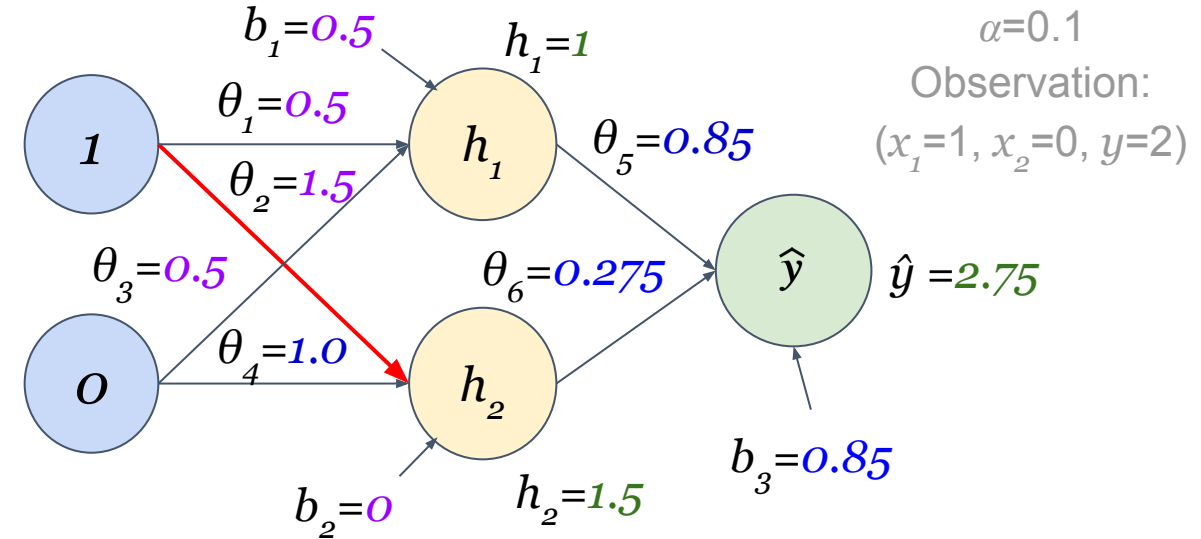
$$\frac{\partial J(\theta)}{\partial h_2} = -2 \cdot \theta_6 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) = 0.75$$

$$\frac{\partial h_2}{\partial \theta_2} = \frac{\partial (f(b_2 + \theta_2 x_1 + \theta_4 x_2))}{\partial \theta_2}$$

$$\frac{\partial h_2}{\partial \theta_2} = \frac{\partial h_2}{\partial z} \cdot \frac{\partial z}{\partial \theta_2} \Rightarrow \frac{\partial h_2}{\partial z} = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases} \text{ and } \frac{\partial z}{\partial \theta_2} = \frac{\partial (b_2 + \theta_2 x_1 + \theta_4 x_2)}{\partial \theta_2} = x_1 = 1$$

Again: $b_2 + \theta_2 x_1 + \theta_4 x_2 = 1.5 > 0 \Rightarrow \frac{\partial h_2}{\partial z} = 1.5$ So: $\frac{\partial J(\theta)}{\partial \theta_4} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial \theta_4} = 0.75 \cdot 1 \cdot 1.5 = 1.125$

$$\begin{aligned} h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75 \end{aligned}$$



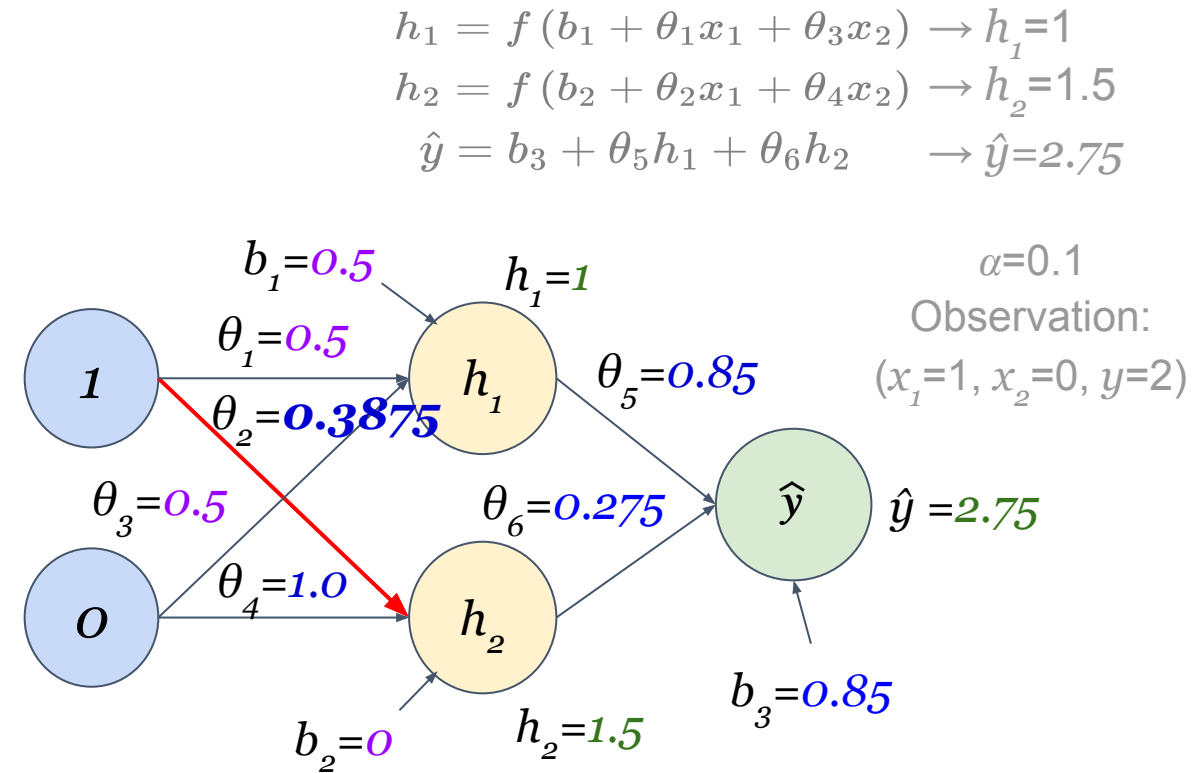
Simple Example

We can do the same thing for θ_2 .

$$\frac{\partial J(\theta)}{\partial \theta_4} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial \theta_4} = 1.125$$

Update θ_2 :

$$\begin{aligned} \theta_2 &= \theta_2 - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_2} \\ &= 0.5 - 0.1 \cdot 1.125 = 0.5 - 0.1125 = 0.3875 \end{aligned}$$



Simple Example

Now θ_3 .

$$\frac{\partial J(\theta)}{\partial \theta_3} = \frac{\partial J(\theta)}{\partial h_1} \cdot \frac{\partial h_1}{\partial \theta_3}$$

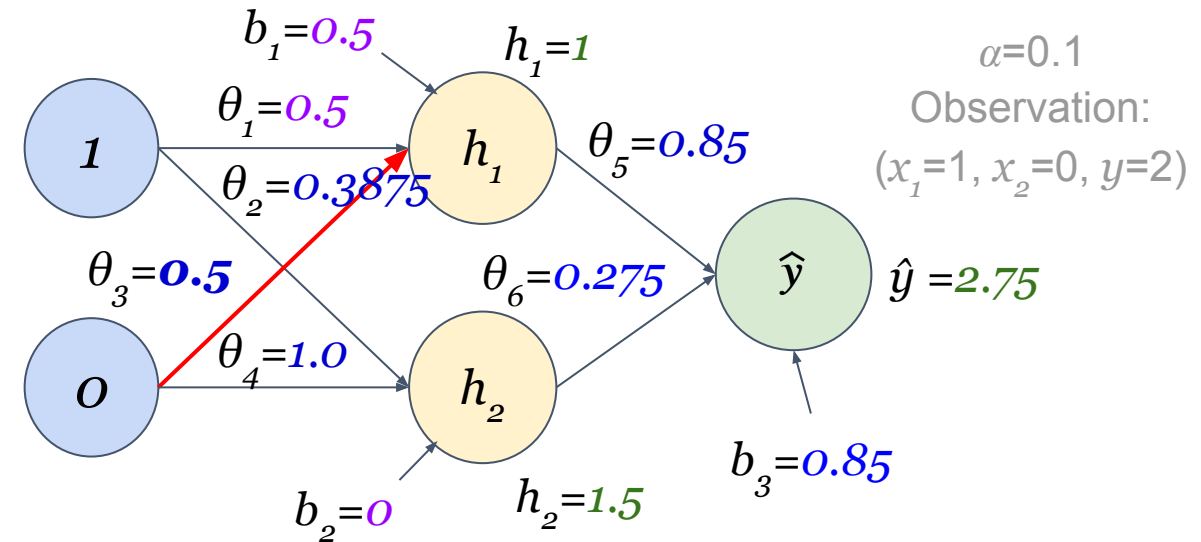
$$\frac{\partial J(\theta)}{\partial h_1} = -2 \cdot \theta_5 (y - b_3 - \theta_5 h_1 - \theta_6 h_2)$$

$$= -2 \cdot 0.5 (2 - 1 - 0.5 \cdot 1 - 0.5 \cdot 1.5) = 1 \cdot (1 - 0.5 - 0.75) = 0.25$$

$$\frac{\partial h_1}{\partial \theta_3} = \frac{\partial (f(b_1 + \theta_1 x_1 + \theta_3 x_2))}{\partial \theta_3}$$

$$\frac{\partial h_1}{\partial \theta_3} = \frac{\partial h_1}{\partial z} \cdot \frac{\partial z}{\partial \theta_3} \Rightarrow \frac{\partial h_1}{\partial z} = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases} \text{ and } \frac{\partial z}{\partial \theta_3} = \frac{\partial (b_1 + \theta_1 x_1 + \theta_3 x_2)}{\partial \theta_3} = x_2 = 0 \rightarrow \text{no change}$$

$$\begin{aligned} h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75 \end{aligned}$$



Simple Example

Now θ_1 .

$$\frac{\partial J(\theta)}{\partial \theta_1} = \frac{\partial J(\theta)}{\partial h_1} \cdot \frac{\partial h_1}{\partial \theta_1}$$

$$\frac{\partial J(\theta)}{\partial h_1} = -2 \cdot \theta_5 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) = 0.25$$

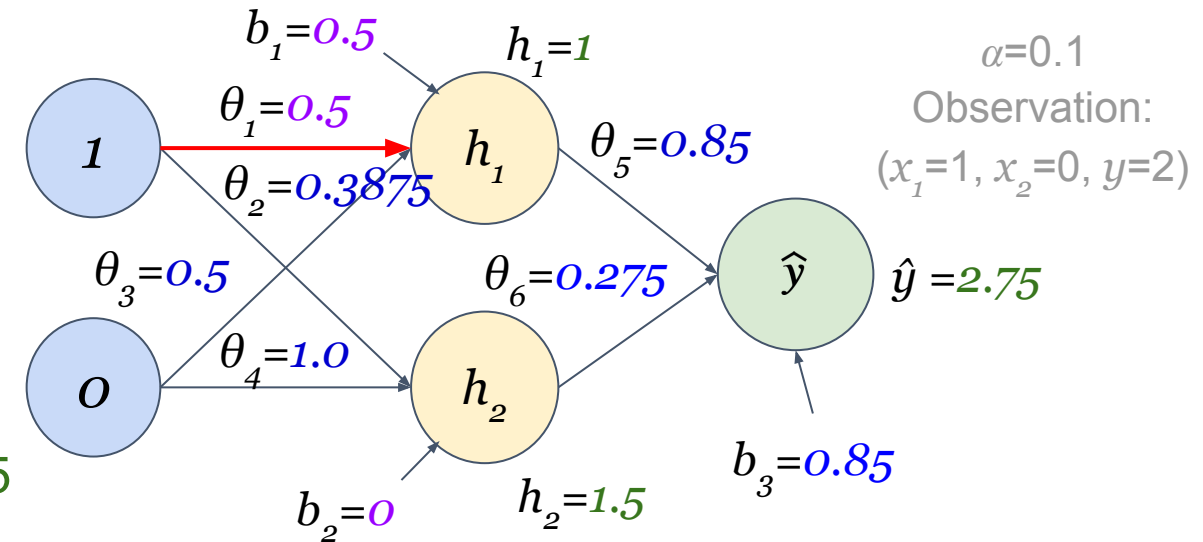
$$\frac{\partial h_1}{\partial \theta_1} = \frac{\partial (f(b_1 + \theta_1 x_1 + \theta_3 x_2))}{\partial \theta_1}$$

$$\frac{\partial h_1}{\partial \theta_1} = \frac{\partial h_1}{\partial z} \cdot \frac{\partial z}{\partial \theta_1} \Rightarrow \frac{\partial h_1}{\partial z} = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases} \text{ and } \frac{\partial z}{\partial \theta_1} = \frac{\partial (b_1 + \theta_1 x_1 + \theta_3 x_2)}{\partial \theta_1} = x_1 = 1$$

$$\text{Now: } b_1 + \theta_1 x_1 + \theta_3 x_2 = 0.5 + 0.5 \cdot 1 + 0.5 \cdot 0 = 1 > 0 \Rightarrow \frac{\partial h_1}{\partial z} = \begin{cases} 1 & \text{if } b_1 + \theta_1 x_1 + \theta_3 x_2 \geq 0 \\ 0 & \text{otherwise} \end{cases} = 1$$

$$\text{So: } \frac{\partial J(\theta)}{\partial \theta_1} = \frac{\partial J(\theta)}{\partial h_1} \cdot \frac{\partial h_1}{\partial \theta_1} = 0.25 \cdot 1 \cdot 1 = 0.25$$

$$\begin{aligned} h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75 \end{aligned}$$



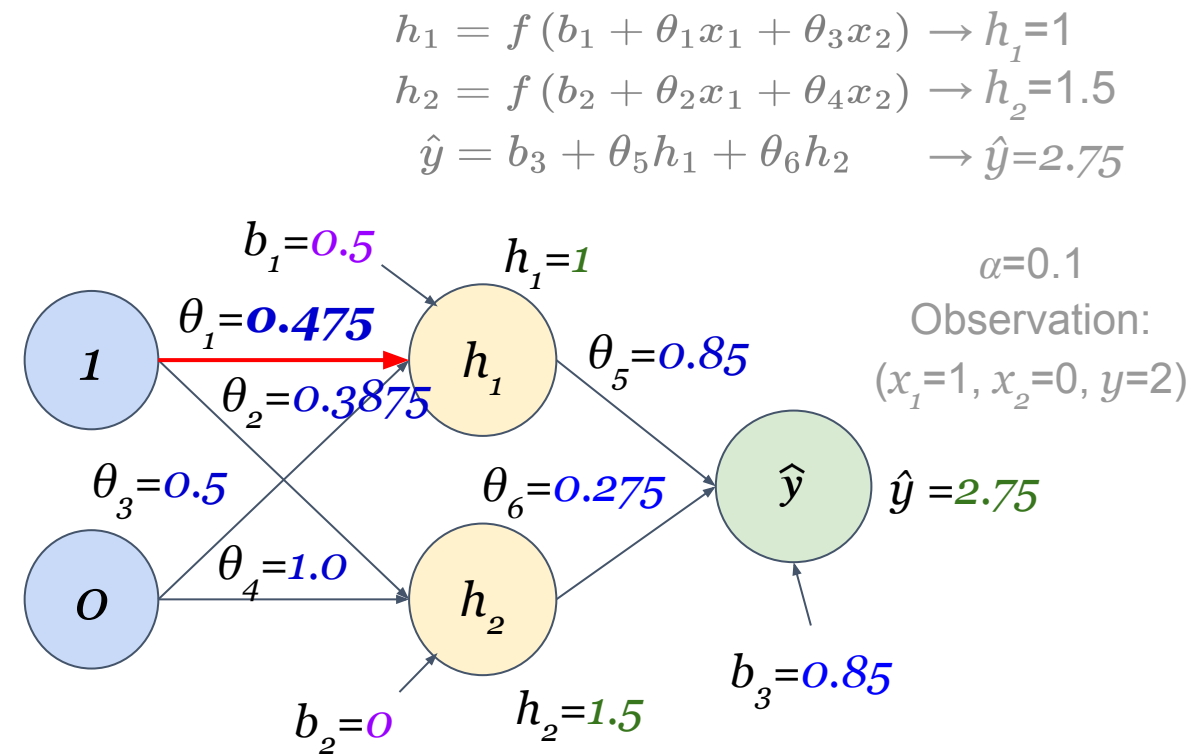
Simple Example

Now θ_1 .

$$\frac{\partial J(\theta)}{\partial \theta_1} = \frac{\partial J(\theta)}{\partial h_1} \cdot \frac{\partial h_1}{\partial \theta_1} = 0.25$$

Update θ_1 :

$$\begin{aligned} \theta_1 &= \theta_1 - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta_1} \\ &= 0.5 - 0.1 \cdot 0.25 = 0.5 - 0.025 = 0.475 \end{aligned}$$



Simple Example

Now b_1 .

$$\frac{\partial J(\theta)}{\partial b_1} = \frac{\partial J(\theta)}{\partial h_1} \cdot \frac{\partial h_1}{\partial b_1}$$

$$\frac{\partial J(\theta)}{\partial h_1} = -2 \cdot \theta_5 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) = 0.25$$

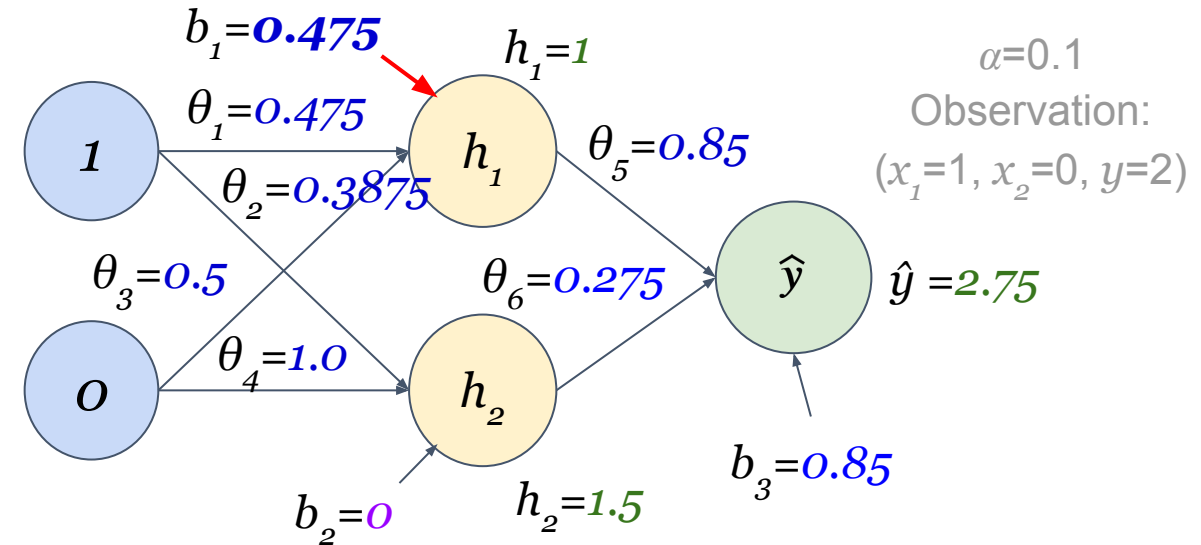
$$\frac{\partial h_1}{\partial b_1} = \frac{\partial (f(b_1 + \theta_1 x_1 + \theta_3 x_2))}{\partial b_1}$$

$$\frac{\partial h_1}{\partial b_1} = \frac{\partial h_1}{\partial z} \cdot \frac{\partial z}{\partial b_1} \Rightarrow \frac{\partial h_1}{\partial z} = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \frac{\partial z}{\partial b_1} = \frac{\partial (b_1 + \theta_1 x_1 + \theta_3 x_2)}{\partial b_1} = 1 = 1$$

$$\text{And: } b_1 + \theta_1 x_1 + \theta_3 x_2 = 1 > 0 \Rightarrow \frac{\partial h_1}{\partial z} = 1$$

$$\text{So: } \frac{\partial J}{\partial b_1} = \frac{\partial J(\theta)}{\partial h_1} \cdot \frac{\partial h_1}{\partial b_1} = 0.25 \cdot 1 \cdot 1 = 0.25$$

$$\begin{aligned} h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75 \end{aligned}$$



$$\begin{aligned} \text{Update } b_1: \quad b_1 &= b_1 - \alpha \cdot \frac{\partial J(\theta)}{\partial b_1} \\ &= 0.5 - 0.1 \cdot 0.25 = 0.5 - 0.025 = 0.475 \end{aligned}$$

Simple Example

Now b_2 .

$$\frac{\partial J(\theta)}{\partial b_2} = \frac{\partial J(\theta)}{\partial h_2} \cdot \frac{\partial h_2}{\partial b_2}$$

$$\frac{\partial J(\theta)}{\partial h_2} = -2 \cdot \theta_6 (y - b_3 - \theta_5 h_1 - \theta_6 h_2) = 0.75$$

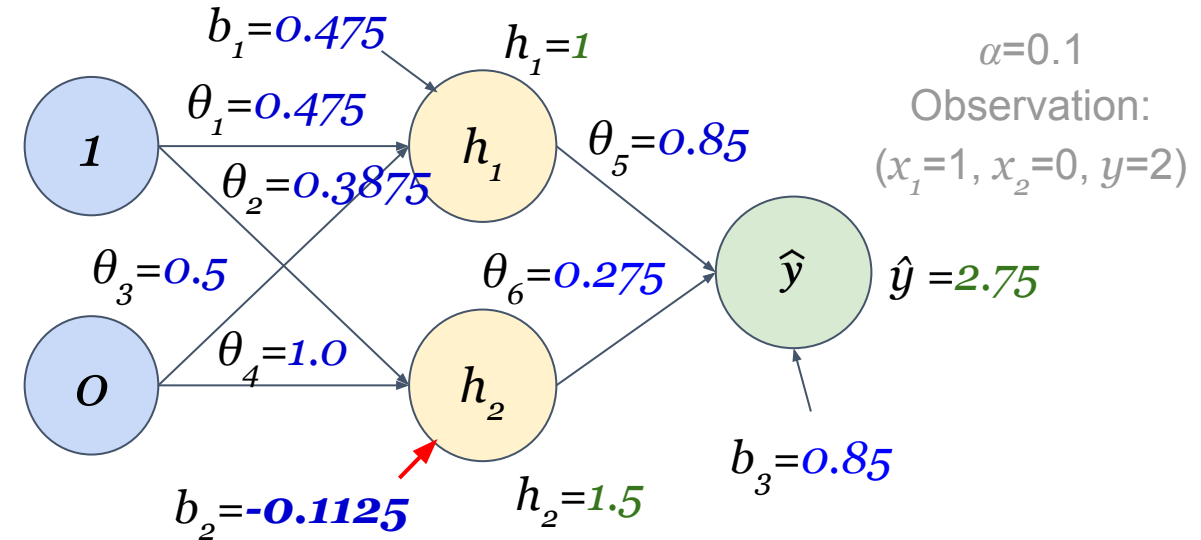
$$\frac{\partial h_2}{\partial b_2} = \frac{\partial (f(b_2 + \theta_2 x_1 + \theta_4 x_2))}{\partial b_2}$$

$$\frac{\partial h_2}{\partial b_2} = \frac{\partial h_2}{\partial z} \cdot \frac{\partial z}{\partial b_2} \Rightarrow \frac{\partial h_2}{\partial z} = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{otherwise} \end{cases} \text{ and } \frac{\partial z}{\partial b_2} = \frac{\partial (b_2 + \theta_2 x_1 + \theta_4 x_2)}{\partial b_2} = 1 = 1$$

$$\text{And: } b_2 + \theta_2 x_1 + \theta_4 x_2 = 1.5 > 0 \Rightarrow \frac{\partial h_2}{\partial z} = 1.5$$

$$\text{So: } \frac{\partial J}{\partial b_1} = \frac{\partial J(\theta)}{\partial h_1} \cdot \frac{\partial h_1}{\partial b_1} = 0.75 * 1.5 * 1 = 1.125$$

$$\begin{aligned} h_1 &= f(b_1 + \theta_1 x_1 + \theta_3 x_2) \rightarrow h_1 = 1 \\ h_2 &= f(b_2 + \theta_2 x_1 + \theta_4 x_2) \rightarrow h_2 = 1.5 \\ \hat{y} &= b_3 + \theta_5 h_1 + \theta_6 h_2 \rightarrow \hat{y} = 2.75 \end{aligned}$$

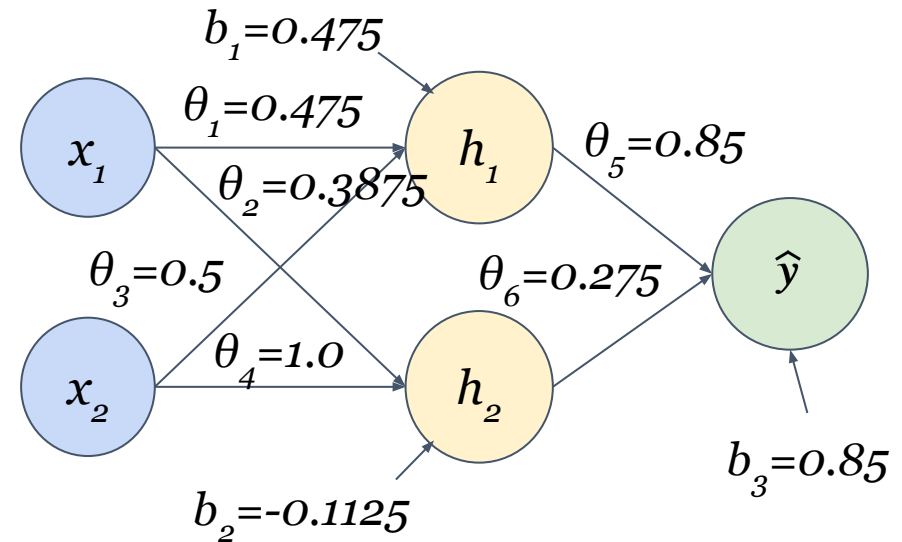


$$\begin{aligned} \text{Update } b_2: \quad b_2 &= b_2 - \alpha \cdot \frac{\partial J(\theta)}{\partial b_2} \\ &= 0 - 0.1 * 1.125 = 0 - 0.1125 = -0.1125 \end{aligned}$$

Simple Example

At this point we've done **one** full forward pass and **one** full backward pass for observation $(1,0,y=2)$.

This process repeats until all training examples have been used.



Bottom line for backpropagation

- The error in prediction for a training example given the current weights is used to adjust the weights
 - We figure out each weight's contribution to the error using math, and update weights that have more responsibility more
- 

Batching

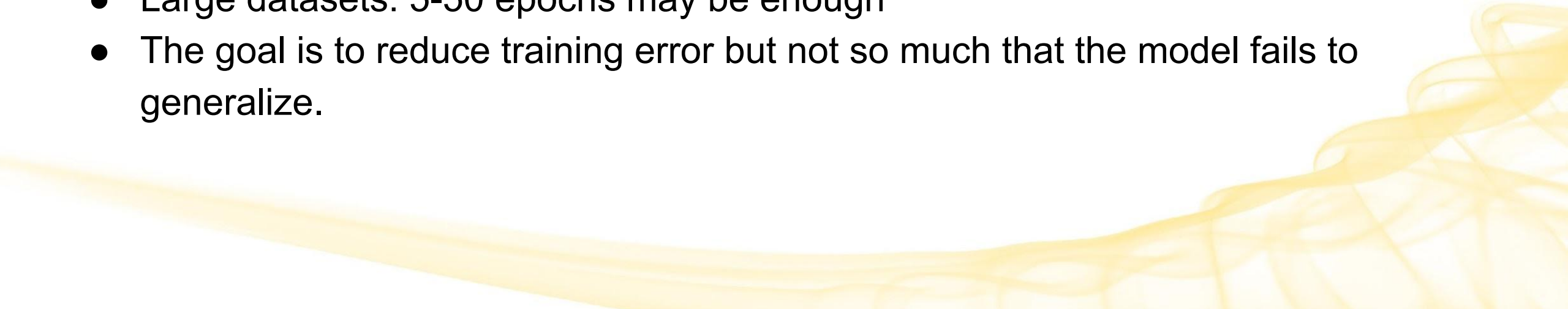
- It's possible to update weights for every training example, like we did
 - Gradient is very noisy → weights jump around a lot
- But you can also split your data into batches of >1 (say 100)
 - Forward pass → compute predictions for batch of examples
 - Compute average loss over the batch
 - Backward pass → gradients computed from batch loss
 - Update weights
 - Provides a good tradeoff between stability and speed
- Or your whole dataset can be a batch
 - Forward pass → compute predictions for all training examples
 - Compute average loss
 - Backprop → update weights

Batching: example

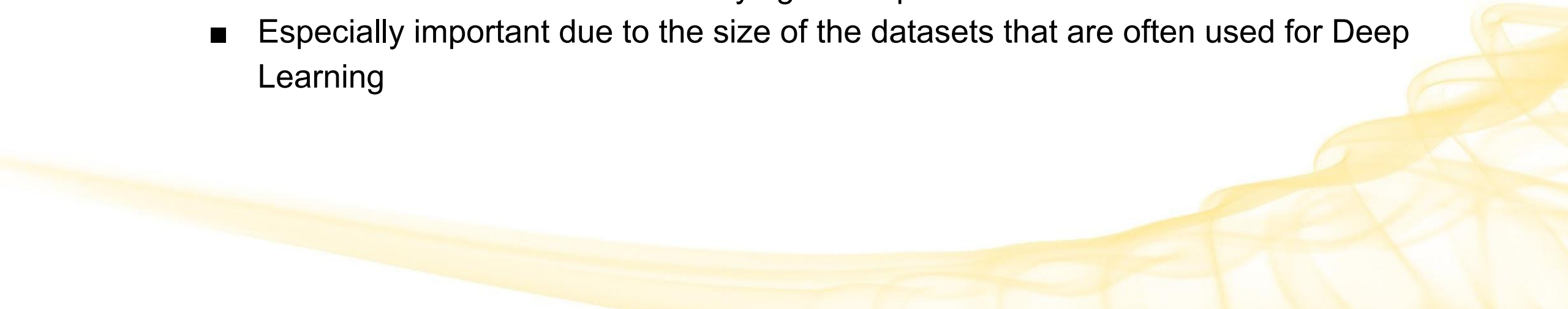
10,000 training examples. You could:

- Compute loss and update weights for every example (update 10,000 times)
 - Split data into 50 groups of 200, and update the weights 50 times based on average loss from each group
 - Update weights once based on average loss from all 10,000 datapoints
- 

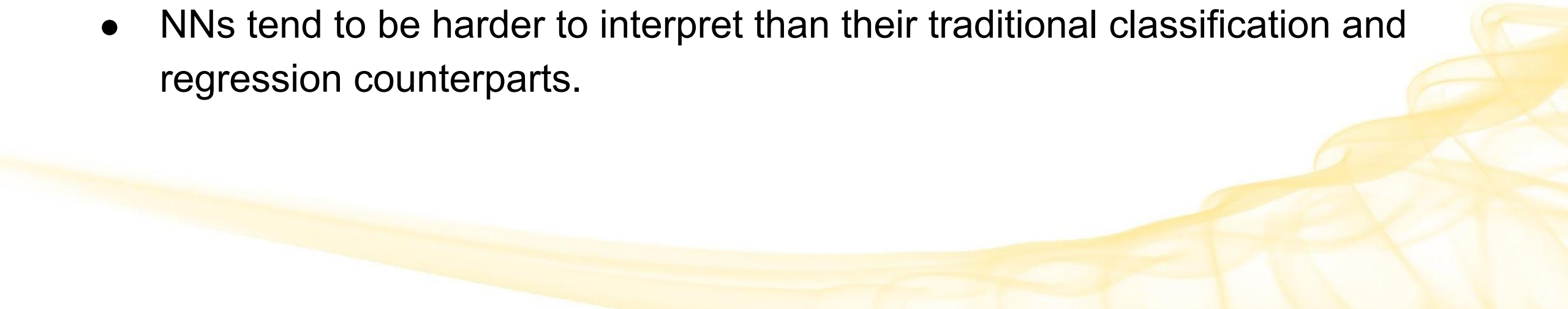
Epochs

- Going over all training examples defines an **epoch**.
 - More epochs → model sees the data more times → can learn more.
 - Too few epochs → underfitting (model hasn't learned enough).
 - Too many epochs → overfitting (model memorizes training data, loses generalization).
 - Small datasets: 50-200 epochs
 - Large datasets: 5-50 epochs may be enough
 - The goal is to reduce training error but not so much that the model fails to generalize.
- 

Multi-Layer Feed-Forward NN Tips

- It's often best to start with a **simple** network, then increase complexity as needed
 - We're talking number of layers and number of nodes per layer
 - Choice of **Activation Function** has major impacts
 - Activation Function selection also impacts processing speed
 - ReLU has a huge reduction in computational costs, but you usually need to use a modified ReLU to deal with the Dying ReLU problem
 - Especially important due to the size of the datasets that are often used for Deep Learning
- 

Multi-Layer Feed-Forward NNs Tips (cont.)

- Multi-layer, feed-forward networks using backpropagation make decent classifiers and regression models.
 - These are almost always supervised
 - They do especially well for unstructured data, like images
 - Somewhat finicky – learning rate can be hard to tune, and the number of layers and hidden nodes typically affects the output.
 - NNs tend to be harder to interpret than their traditional classification and regression counterparts.
- 

Bottom line for NNs

- There are many things to choose when training NNs.
 - We just scratched the surface of what they are.
 - Important to understand how to do a forward pass with numbers, and how backpropagation works conceptually.
- 

On Wednesday

- A simple NN demo
- Data Science/AI Ethics

